# Chapter 2
# Regulations, Maintenance Forms, Records, & Publications

## Overview—Title 14 of the Code of Federal Regulations (14 CFR)

*A*viation-related regulations that have occurred from 1926–1966 are reflected in *Figure 2-1*. Just as aircraft continue to evolve with ever improving technology, so do the regulations, publications, forms, and records required to design, build, and maintain them.

The Federal Aviation Administration (FAA) regulations that govern today's aircraft are found in Title 14 of the Code of Federal Regulations (14 CFR). *[Figure 2-2]* There are five volumes under Title 14, Aeronautics and Space. The first three volumes containing 75 active regulations address the Federal Aviation Administration. The fourth volume deals with the Office of the Secretary of the Department of Transportation (Aviation Proceedings) and Commercial Space Transportation, while the fifth volume addresses the National Aeronautics and Space Administration (NASA) and Air Transportation System Stabilization.

These regulations can be separated into the following three categories:

1. Administrative
2. Airworthiness Certification
3. Airworthiness Operation

Since 1958, these rules have typically been referred to as "FARs," short for Federal Aviation Regulations. However, another set of regulations, Title 48, is titled "Federal Acquisitions Regulations," and this has led to confusion with the use of the acronym "FAR." Therefore, the FAA began to refer to specific regulations by the term "14 CFR part XX." Most regulations and the sections within are odd numbered, because the FAA realized in 1958 when the Civil Aeronautics Regulations were recodified into the Federal Aviation Regulations that it would be necessary to add regulations later.

Over the years, the FAA has sometimes seen the need to issue Special Federal Aviation Regulations (SFAR). *[Figure 2-3]* These are frequently focused very specifically on a unique situation and are usually given a limited length of time for effectiveness. Note that the SFAR number is purely a sequential number and has no relevance to the regulation it is addressing or attached to.

The remainder of this handbook focuses only on those regulations relative to airworthiness certification. There are 30 of these listed in *Figure 2-4*, and they are shown graphically in *Figure 2-5*. A significant benefit of this chart is the visual effect showing the interaction of the regulation with other regulations and the placement of the regulation relative to its impact on airworthiness. It is fundamentally important that the definition of the term "airworthy" be clearly understood.

Only recently did the FAA actually define the term "airworthy" in a regulation. (Refer to the 14 CFR part 3 excerpt following this paragraph.) Prior to this definition in part 3, the term could be implied from reading part 21, section 21.183. The term was defined in other non-regulatory FAA publications, and could also be implied from the text found in block 5 of FAA Form 8100-2, Standard Airworthiness Certificate. This certificate is required to be visibly placed on board each civil aircraft. (Refer to "Forms" presented later in this chapter.)

### Title 14 CFR Part 3—General Requirements

Definitions. The following terms have the stated meanings when used in 14 CFR part 3, section 3.5, Statements about products, parts, appliances and materials.

- Airworthy means the aircraft conforms to its type design and is in a condition for safe operation.
- Product means an aircraft, aircraft engine, or aircraft propeller.
- Record means any writing, drawing, map, recording, tape, film, photograph or other documentary material by which information is preserved or conveyed in any format, including, but not limited to, paper, microfilm, identification plates, stamped marks, bar codes or electronic format, and can either be separate from, attached to or inscribed on any product, part, appliance or material.

Airworthiness can be divided into two areas: original airworthiness as depicted in *Figure 2-5,* and recurrent airworthiness as depicted in *Figure 2-6.* There are three primary regulations that govern the airworthiness of an aircraft:

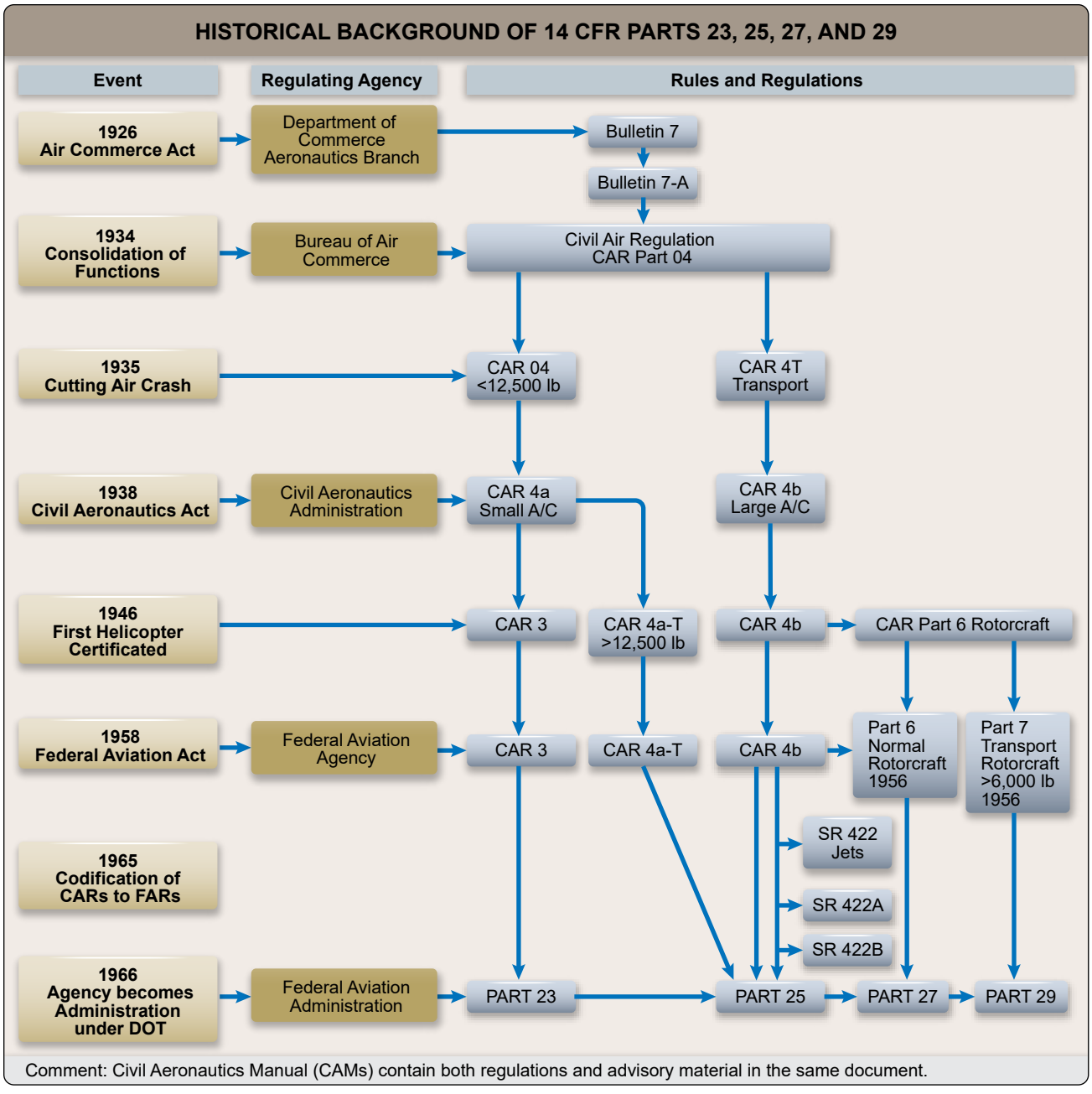



**Figure 2-1.** *FAA historical background of aircraft airworthiness regulations.*

1. 14 CFR part 21—Certification Procedures for Products and Parts
2. 14 CFR part 43—Maintenance, Preventive Maintenance, Rebuilding, and Alterations
3. 14 CFR part 91—General Operating and Flight Rules

Note that the chart in *Figures 2-5* and *2-6* show most of the other airworthiness certification regulations link to one of these regulations.

Although the history section that opens this chapter discusses the FAA as if it was a single unit, it is important to understand that there are various subgroups within the FAA, and each have different responsibilities of oversight in the aviation industry. These may vary by organizational chart or geographic location.

The maintenance technician interacts mostly with FAA personnel from the Flight Standards Service (AFS) and the Flight Standards District Office (FSDO) but may also have some interaction with FAA personnel from the Aircraft Certification Service (AIR).

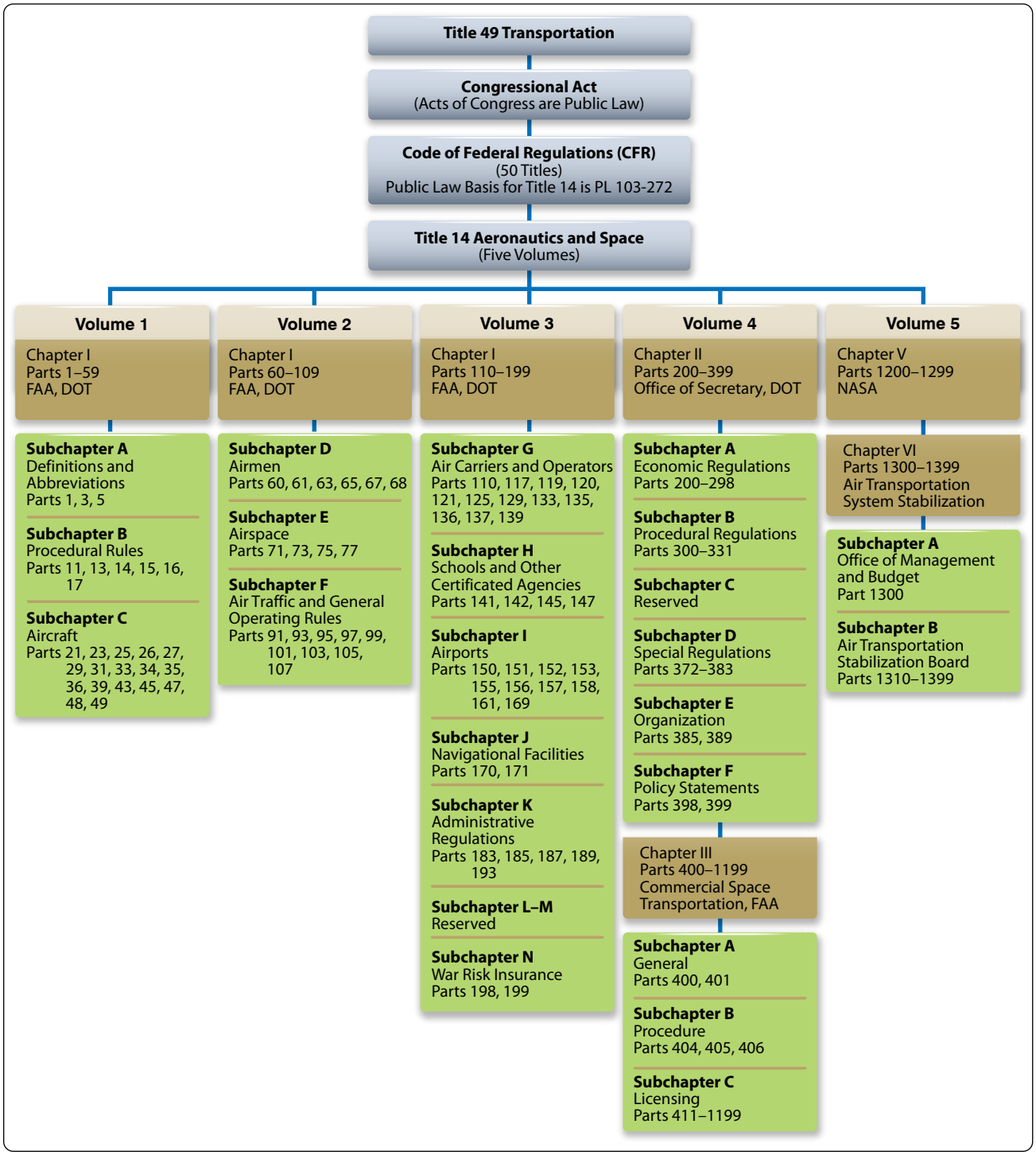



**Figure 2-2.** *Title 14 of the Code of Federal Regulations.*

## Maintenance-Related Regulations

### *14 CFR Part 1—Definitions and Abbreviations*

This section is a very comprehensive, but certainly not all inclusive, list of definitions that both pilots and mechanics must become familiar with. Many regulations often provide additional definitions that are unique to their use and interpretation in that specific part. Title 14 CFR part 1, section 1.2, Abbreviations and Symbols, tends to be highly focused on those abbreviations related to flight.

### *14 CFR Part 21—Certification Procedures for Products and Articles*

This regulation, the first of the three, identifies the requirements of and the procedures for obtaining type

| Special Federal Aviation Regulations | | |
|---|---|---|
| **SFAR No** | **Title** | **Appears in 14 CFR** |
| 13 | Table of Contents | Part 25 |
| 23 | Special Federal Aviation Regulation No. 23 | Part 23 |
| 27-5 | Fuel venting and exhaust emission requirements for turbine engine powered airplanes. | Part 11 |
| 50-2 | Special Flight rules in the Vicinity of the Grand Canyon National Park, AZ | Part 91 |
| 65-1 | Removal of this SFAR effective 10/8/2004 - Prohibition Against Certain Flights Between the United States and Libya | Part 91 |
| 71 | Special Operating Rules for Air Tour Operators in the State of Hawaii | Part 91 |
| 73 | Robinson R-22/R-44 Special Training and Experience Requirements | Part 61 |
| 77 | Prohibition Against Certain Flights Within the Territory and Airspace of Iraq | Part 91 |
| 79 | Prohibition Against Certain Flights Within the Flight Information Region of the Democratic People's Republic of Korea | Part 91 |
| 82 | Prohibition Against Certain Flights within the Territory and Airspace of Sudan | Part 91 |
| 84 | Prohibition Against Certain Flights Within the Territory and Airspace of Serbia-Montenegro | Part 91 |
| 86 | Airspace and Flight Operations Requirement for the Kodak Albuquerque International Balloon Fiesta; Albuquerque, NM | Part 91 |
| 88 | Fuel Tank System Fault Tolerance Evaluation Requirements | Parts 21, 25, 91, 121, 125, 129 |
| 97 | Special Federal Aviation Regulation No. 97 - Special Operating Rules for the Conduct of Instrument Flight Rules (IFR) Area Navigation (RNAV) Operations using Global Positioning Systems (GPS) in Alaska | Parts 71, 91, 95, 121, 125, 129, 135 |
| 98 | Construction or Alteration in the Vicinity of the Private Residence of the President of the United States | Part 77 |
| 100-2 | Relief for U.S. Military and Civilian Personnel Who Are Assigned Outside the United States in Support of U.S. Armed Forces Operations | Parts 61, 63, 65 |
| 103 | Process for Requesting Waiver of Mandatory Separation Age for Certain Federal Aviation Administration (FAA) Air Traffic Control Specialists | Part 65 |
| 104 | Prohibition Against Certain Flights by Syrian Air Carriers to the United States | Part 91 |
| 108 | Special Federal Aviation Regulation No. 108 - Mitsubishi MU-2B Series Airplane Special Training, Experience, and Operating Requirements | Part 91 |
| 109 | Special Requirements for Private Use Transport Category Airplanes | Parts 21, 25, 119 |
| 112 | Prohibition Against Certain Flights Within the Tripoli (HLLL) Flight Information Region (FIR) | Part 91 |

**Figure 2-3.** *Special Federal Aviation Regulations From 14 CFR.*

certificates (TCs), supplemental type certificates (STCs), production certificates, airworthiness certificates, and import and export approvals. *[Figure 2-5]* Some of the other major areas covered in this part are the procedures for obtaining a Part Manufacture Approval (PMA) or an authorization related to producing a Technical Standard Order (TSO) part. Note that part 21's greatest significance is in the original airworthiness phase, although it has minor application in recurrent airworthiness. *[Figure 2-5]* One of the most important sections of this regulation is section 21.50, "Instructions for continued airworthiness and manufacturer's maintenance manuals having airworthiness limitations sections." When an aircraft is delivered new from the manufacturer, it comes with maintenance manuals that define the inspection and maintenance actions necessary to maintain the aircraft in airworthy condition. Also, any STC modification that was developed after 1981 must have, as part of the STC documentation, a complete set of instructions for continued airworthiness (ICA). This ICA contains inspection and maintenance information intended to be used by the technician in maintaining that part of the aircraft that has been altered since it was new. This ICA is comprised of 16 specific subjects. *[Figure 2-7]* An ICA developed in accordance with this checklist should be acceptable to the Aviation Safety Inspector (ASI) reviewing a major alteration.

### *14 CFR Part 23—Airworthiness Standards: Normal, Utility, Acrobatic, and Commuter Category Airplanes*

Aircraft certificated under 14 CFR part 23 represent the greatest portion of what the industry refers to as "general aviation." These aircraft vary from the small two-place piston engine, propeller-driven trainers that are frequently used for flight training, to turbine-powered corporate jets used to transport business executives. Seating capacity is limited to nine or less on all aircraft, except the commuter aircraft where the maximum passenger seating is 19, excluding the pilot and copilot seats.

This part specifies the airworthiness standards that must

| | |
|---|---|
| Part 1 | Definitions and Abbreviations |
| Part 13 | Investigative and Enforcement Procedures |
| Part 21 | Certification Procedures for Products and Parts |
| Part 23 | Airworthiness Standards: Normal, Utility, Acrobatic, and Commuter Category Airplanes |
| Part 25 | Airworthiness Standards: Transport Category Airplanes |
| Part 27 | Airworthiness Standards: Normal Category Rotorcraft |
| Part 29 | Airworthiness Standards: Transport Category Rotorcraft |
| Part 31 | Airworthiness Standards: Manned Free Balloons |
| Part 33 | Airworthiness Standard: Aircraft Engines |
| Part 34 | Fuel Venting and Exhaust Emission Requirements for Turbine Engine Powered Airplanes |
| Part 35 | Airworthiness Standards: Propellers |
| Part 36 | Noise Standards: Aircraft Type and Airworthiness Certification |
| Part 39 | Airworthiness Directives |
| Part 43 | Maintenance, Preventive Maintenance, Rebuilding, and Alteration |
| Part 45 | Identification and Registration Marking |
| Part 47 | Aircraft Registration |
| Part 48 | Registration and Marking Requirements for Small Unmanned Aircraft |
| Part 61 | Certification: Pilots, Flight Instructors, and Ground Instructors |
| Part 63 | Certification: Flight Crewmembers Other Than Pilots |
| Part 65 | Certification: Airmen Other Than Flight Crewmembers |
| Part 68 | Requirements for Operating Certain Small Aircraft Without a Medical Certificate |
| Part 91 | General Operating and Flight Rules |
| Part 107 | Small Unmanned Aircraft Systems |
| Part 119 | Certification: Air Carriers and Commercial Operators |
| Part 121 | Operating Requirements: Domestic, Flag, and Supplemental Operations |
| Part 125 | Certification and Operations: Airplanes Having a Seating Capacity of 20 or More Passengers or a Maximum Payload Capacity of 6,000 Pounds or More; and Rules Governing Persons on Board Such Aircraft |
| Part 135 | Operating Requirements: Commuter and On Demand Operations and Rules Governing Persons on Board Such Aircraft |
| Part 145 | Repair Stations |
| Part 147 | Aviation Maintenance Technician Schools |
| Part 183 | Representatives of the Administrator |

**Figure 2-4.** *List of FAA regulations relative to airworthiness certification.*

be met in order for a manufacturer to receive a TC and for the aircraft to receive an airworthiness certificate. Part 23 aircraft are those aircraft that have a maximum certificated takeoff weight of 12,500 pounds or less, except for those aircraft in the commuter category. The maximum certificated takeoff weight limit rises to 19,000 pounds or less for these aircraft.

Part 23 has seven subparts, six of them providing detailed criteria for the design of these aircraft. The first, subpart A, defines the applicability of this regulation. The others are:

- Subpart B—Flight
- Subpart C—Structures
- Subpart D—Design and Construction
- Subpart E—Powerplant
- Subpart F—Equipment
- Subpart G—Flightcrew Interface and Other Information

Within each of these subparts are numerous sections that specify details, such as center of gravity (CG), gust load factors, removable fasteners, the shape of certain flight deck controls, engine and propeller requirements, fuel tank markings, flight deck instrumentation marking and placards, cabin aisle width, and flammability resistant standards.

### *14 CFR Part 25—Airworthiness Standards: Transport Category Airplanes*

The standards in 14 CFR part 25 apply to large aircraft with a maximum certificated takeoff weight of more than 12,500 pounds. This segment of aviation is usually referred to as "commercial aviation" and includes most of the aircraft seen at a large passenger airport, except for the commuter aircraft included in 14 CFR part 23. However, the ability to carry passengers is not a requirement for aircraft certified to 14 CFR part 25. Many of these aircraft are also used to transport cargo. This chapter is subdivided into similar design subpart categories and the same sequence as the requirements specified in 14 CFR part 23.

### *14 CFR Part 27—Airworthiness Standards: Normal Category Rotorcraft*

This regulation deals with the small rotor wing aircraft and is consistent with 14 CFR part 23 with limiting the passenger seating to nine or less. However, the maximum certificated weight is limited to 7,000 pounds. It contains similar design subparts identified in 14 CFR part 23 that provide the details for designing the aircraft.

### *14 CFR Part 29—Airworthiness Standards: Transport Category Rotorcraft*

This section specifies those standards applicable to helicopters with a maximum certified weight greater than 7,000 pounds. However, it also includes additional parameters based upon seating capacity and an additional weight limit. Those parameters are passenger seating, (nine or less, ten or more) and whether the helicopter is over or under a maximum weight of 20,000 pounds. The design subparts of part 29 are similar to those in 14 CFR parts 23, 25, and 27.

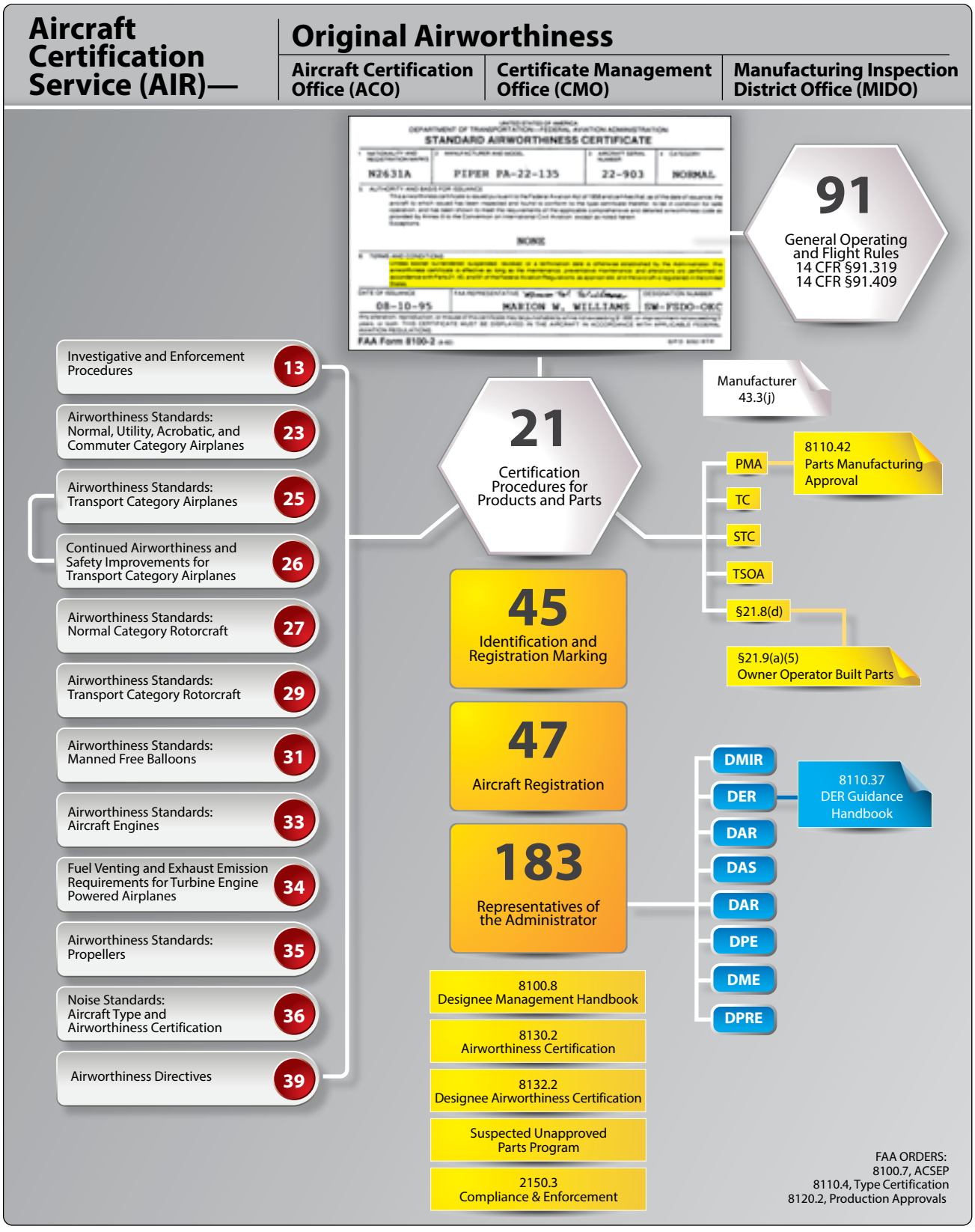



**Figure 2-5.** *Graphic chart of FAA regulations.*

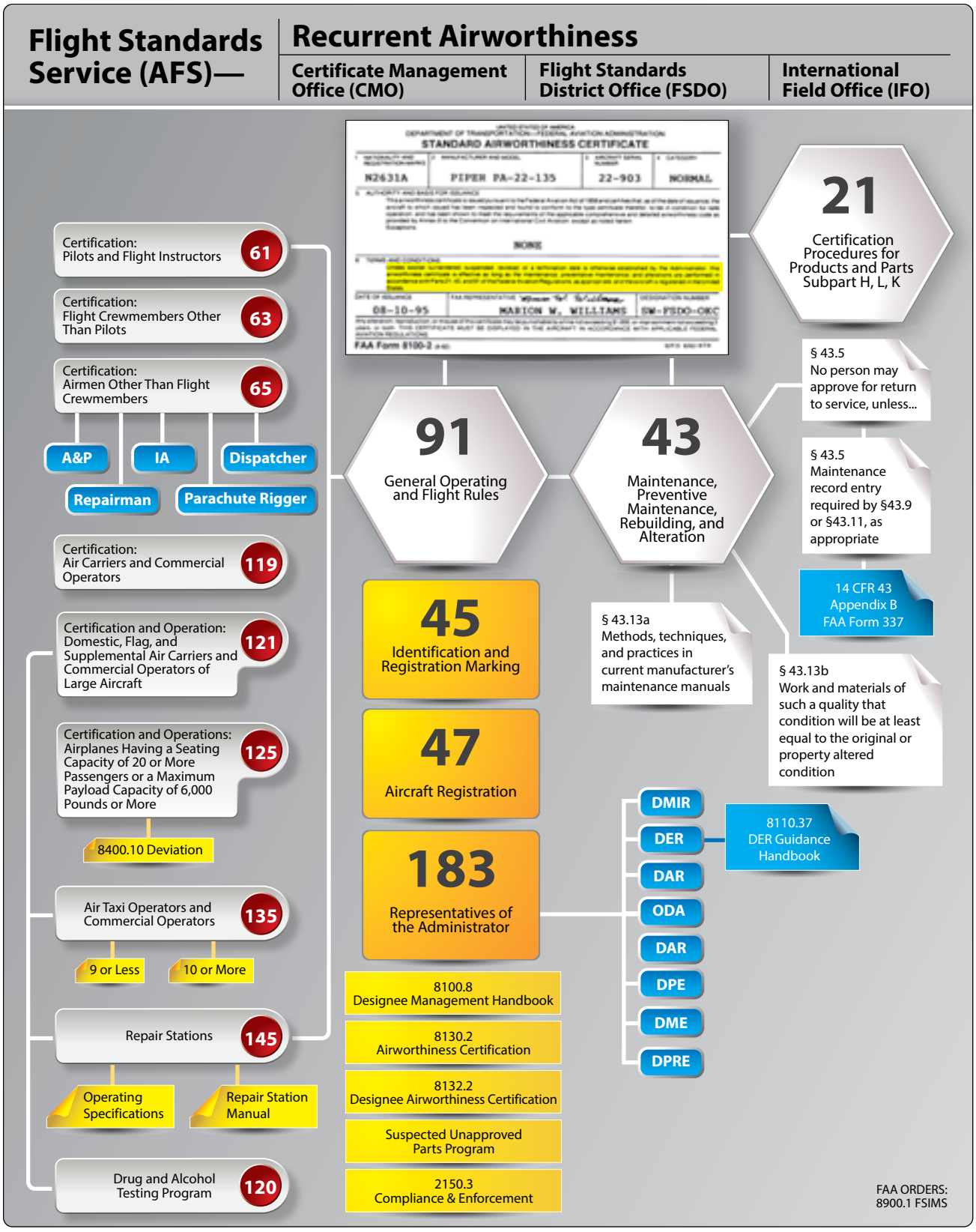



**Figure 2-6.** *Graphic chart of FAA regulations (continued).*

### *14 CFR Part 33—Airworthiness Standards: Aircraft Engines*

Each of the four preceding 14 CFR regulations require that the engine used in the aircraft must be type certificated. Title 14 CFR part 33 details the requirements for both reciprocating and turbine style aircraft engines. It not only specifies the design and construction requirements, but also the block test requirements that subject the engine to extremely demanding testing in order to prove its capability of enduring the stresses of powering the aircraft.

### *14 CFR Part 35—Airworthiness Standards: Propellers*

Just as each engine used on an aircraft must have a TC, the propeller must also be type certificated. This part is arranged the same way that 14 CFR part 33 is, in that subpart B specifies design and construction while subpart C covers tests and inspections.

Since regulations change over the years, not every aircraft presently flying meets the current design regulations as printed this year. When regulations are revised, they are printed in the Federal Register and released with an amendment number that ties them to the regulation being revised. Aircraft are required to meet only the specifications in force at the time the aircraft is built. ***Note:*** The preceding statement does not apply to the mandatory requirements imposed by Airworthiness Directives (AD), as these usually have a compliance date included in the AD note.

### *14 CFR Part 39—Airworthiness Directives*

In spite of all the emphasis on proper design and certification testing, sometimes the actual day-to-day use of the aircraft causes unanticipated wear or failure to occur. When that happens, if the FAA determines that the wear or failure represents an unsafe condition and that the condition is likely to exist in other products of the same type of design, it issues an AD. Actual AD notes are not included in 14 CFR part 39, but rather are printed in the Federal Register and are linked to this part as amendments to 14 CFR part 39, section 39.13. AD notes are legally enforceable rules that apply to aircraft, aircraft engines, propellers, and appliances.

### *14 CFR Part 43—Maintenance, Preventive Maintenance, Rebuilding, and Alteration*

This regulation represents the heart of aviation maintenance and is one of the three major regulations previously identified. The 13 rules and 6 appendices contained within 14 CFR part 43 provide the standard for maintaining all civilian aircraft currently registered in the United States. Note that 14 CFR part 43 has a significant relationship with part 91 and other parts in maintaining continued airworthiness. *[Figure 2-6]* A more detailed explanation of this regulation is presented later in this handbook.

### *14 CFR Part 45—Identification and Registration Marking*

Title 14 of the CFR part 45 includes the requirements for the identification of aircraft, engines, propellers, certain replacement and modification parts, and the nationality and registration marking required on U.S.-registered aircraft. All type-certificated products must have the following information on a fireproof dataplate or similar approved fireproof method.

1. Builder's name
2. Model designation
3. Builder's serial number
4. TC number (if any)
5. Production certificate number (if any)
6. For aircraft engines, the established rating
7. Reference to compliance or exemption to 14 CFR Part 34, Fuel Venting and Exhaust Emission Requirements for Turbine Engine Powered Airplanes
8. Any other information that the FAA determines to be appropriate

Replacement and modification parts are produced in accordance with a Parts Manufacturer Approvals (PMA) (14 CFR part 21, section 21.303) and must have the following information permanently and legibly marked:

1. The letters "FAA-PMA"
2. The name, symbol, or trademark of the holder of the PMA
3. The part number
4. The name and model designation for each type certificated product it can be installed on

If a part has a specified replacement time, inspection interval, or other related procedure specification in the maintenance manual or ICA, that part must have a part number and a serial number (or the equivalent of each).

The manufacturer of a life-limited part must either provide marking instructions for that part, or state that the part cannot be marked without a compromise to its integrity. Exceptions are made for the identification of parts that are too small to be practical to mark the required data.

Nationality and registration marks (commonly known as the N-number for U.S.-registered aircraft) can vary in size, depending on the year that the aircraft was built and whether or not the aircraft has been repainted. The most common size is at least 12 inches in height. Small aircraft built at least 30

**ITEM SUBJECT**

1. **Introduction:** Briefly describes the aircraft, engine, propeller, or component that has been altered. Include any other information regarding the content, scope, purpose, arrangement, applicability, definitions, abbreviations, precautions, units of measurement, list of parts used, referenced publications, and distribution of the ICA, as applicable.
2. **Description:** Of the major alteration and its functions, including an explanation of its interface with other systems, if any.
3. **Control, operation information:** Or special procedures, if any.
4. **Servicing information:** Such as types of fluids used, servicing points, and location of access panels, as appropriate.
5. **Maintenance instructions:** Such as recommended inspection/maintenance periods in which each of the major alteration components are inspected, cleaned, lubricated, adjusted, and tested, including applicable wear tolerances and work recommended at each scheduled maintenance period. This section can refer to the manufacturers' instructions for the equipment installed where appropriate (e.g., functional checks, repairs, inspections). It should also include any special notes, cautions, or warnings, as applicable.
6. **Troubleshooting information:** Describes probable malfunctions, how to recognize those malfunctions, and the remedial actions to take.
7. **Removal and replacement information:** Describes the order and method of removing and replacing products or parts, and any necessary precautions. This section should also describe or refer to manufacturer's instructions to make required tests, checks, alignment, calibrations, center of gravity changes, lifting, or shoring, etc., if any.
8. **Diagrams:** Of access plates and information, if needed, to gain access for inspection.
9. **Special inspection requirements:** Such as X-ray, ultrasonic testing, or magnetic particle inspection, if required.
10. **Application of protective treatments:** To the affected area after inspection and/or maintenance, if any.
11. **Data:** Relative to structural fasteners such as type, torque, and installation requirements, if any.
12. **List of special tools:** Special tools that are required, if any.
13. **For commuter category aircraft:** Provide the following additional information, as applicable:
    A. Electrical loads
    B. Methods of balancing flight controls
    C. Identification of primary and secondary structures
    D. Special repair methods applicable to the aircraft
14. **Recommended overhaul periods:** Required to be noted on the ICA when an overhaul period has been established by the manufacturer of a component or equipment. If no overhaul period exists, the ICA should state for item 14, "No additional overhaul time limitations."
15. **Airworthiness limitation section:** Includes any "approved" airworthiness limitations identified by the manufacturer or FAA Type Certificate Holding Office (e.g., An STC incorporated in a larger field-approved major alteration may have an airworthiness limitation). The FAA inspector should not establish, alter, or cancel airworthiness limitations without coordinating with the appropriate FAA Type Certificate Holding Office. If no changes are made to the airworthiness limitations, the ICA should state for item 15, "No additional airworthiness limitations" or "Not Applicable."
16. **Revision:** Includes information on how to revise the ICA. For example, a letter will be submitted to the local FAA Office with a copy of the revised FAA Form 337 and revised ICA. The FAA inspector accepts the change by signing block 3 and including the following statement: "The attached revised/new Instructions for Continued Airworthiness (date_____) for the above aircraft or component major alteration have been accepted by the FAA, superseding the Instructions for Continued Airworthiness (date_____)." After the revision has been accepted, a maintenance record entry will be made, identifying the revision, its location, and date on the FAA Form 337.

SAMPLE

**Figure 2-7.** *Instructions for Continued Airworthiness (ICA) Checklist.*

years ago, or replicas of these, or experimental exhibition or amateur-built aircraft may use letters at least 2 inches in height. Only a few aircraft are authorized to display registration markings of at least 3 inches. Note that this regulation sits directly on the vertical line in *Figure 2-5* indicating that it applies to both original and recurrent airworthiness.

### *14 CFR Part 47—Aircraft Registration*

This regulation provides the requirements for registering aircraft. It includes procedures for both owner and dealer registration of aircraft.

### *14 CFR Part 65—Certification: Airmen Other Than Flight Crewmembers*

Pilots, flight instructors, and ground instructors are certificated under 14 CFR part 61. Flight crew other than pilots are certificated under 14 CFR part 63. However, many other people are also required to be certificated by the FAA for the U.S. aviation fleet to operate smoothly and efficiently. Title 14 CFR part 65 addresses many of those other people.

- Subpart B—Air Traffic Control Tower Operators
- Subpart C—Aircraft Dispatchers
- Subpart D—Mechanics
- Subpart E—Repairmen
- Subpart F—Parachute Riggers

A more detailed discussion of this chapter with a special emphasis on mechanics is included in Chapter 15, The Mechanic Certificate.

***Note:*** SFAR 100-2. Relief for U.S. Military and Civilian Personnel who are assigned outside the United States in support of U.S. Armed Forces Operations is a good example of the specific nature and limited time frame that are part of a SFAR.

### *14 CFR Part 91—General Operating and Flight Rules*

This is the final regulation of the three major regulations identified earlier in this chapter. Note its interaction in *Figure 2-6* with other regulations visually indicating its "operational" involvement or "recurrent airworthiness." Although it is an operational regulation that is focused toward the owner, operator, and/or pilot of the aircraft, the maintenance technician must have an awareness of this regulation. Two examples of these maintenance related issues are:

1. Section 91.207—Emergency Locator Transmitters

   Paragraph (c)(2)—battery replacement interval and requirement for a logbook entry indicating the expiration date of the new battery.

2. Section 91.213—Inoperative Instruments and Equipment

   Paragraph (a)(2)—a letter of authorization from the FSDO authorizing the operation of the aircraft under a Minimum Equipment List (MEL) constitutes a STC and must be carried in the aircraft during flight.

*Subpart E—Maintenance, Preventive Maintenance, and Alterations (Sections 91.401 through 91.421)*

This is the section of most interest to the technician. They must be familiar with it, because it does carry some (indirect) responsibility for the technician. Note that the 14 CFR part 91 icon in *Figure 2-6* has a direct line to 14 CFR part 43. This is because section 91.403(b) states, "No person may perform maintenance, preventive maintenance, or alterations on an aircraft other than as prescribed in this subpart and other applicable regulations, including part 43 of this chapter." A more complete discussion of this regulation, especially Subpart E—Maintenance, Preventive Maintenance, and Alterations is presented later in this chapter.

### *14 CFR Part 119—Certification: Air Carriers and Commercial Operators*

In order to better understand the next three regulations discussed here (14 CFR parts 121, 125, and 135) a brief overview of 14 CFR part 119 is beneficial. [*Figure 2-8]* There are more than 50 Advisory Circulars (ACs) in the 120 series alone providing additional non-regulatory information concerning the variety of procedures involved with these operations. There are basically three different criteria that must be analyzed in order to properly determine the regulation that applies. These are:

1. Is the service provided for Private Carriage or Common Carriage?
2. Is the aircraft For Hire or is it Not for Hire?
3. Is it a large or small aircraft?

AC 120-12, as revised, provides the following definition regarding this criterion: A carrier becomes a common carrier when it "holds itself out" to the public, or to a segment of the public, as willing to furnish transportation within the limits of its facilities to any person who wants it. There are four elements in defining a common carrier:

1. A holding out of a willingness to
2. Transport persons or property
3. From place to place
4. For compensation

This "holding out" that makes a person a common carrier can be done in many ways, and it does not matter how it is done. Signs and advertising are the most direct means of "holding out," but are not the only ones.

Carriage for hire which does not involve "holding out" is private carriage. Private carriers for hire are sometimes called "contract carriers," but the term is borrowed from the Interstate Commerce Act and legally inaccurate when used in connection with the Federal Aviation Act. Private carriage for hire is carriage for one or several selected customers, generally on a long-term basis. The number of contracts must not be too great; otherwise, it implies a willingness to make a contract with anybody. A carrier operating pursuant to 18 to 24 contracts has been held to be a common carrier, because it held itself out to serve the public generally to the extent of its facilities. Private carriage has been found in cases where three contracts have been the sole basis of the operator's business.

Operations that constitute common carriage are required to be conducted under 14 CFR part 121 or 135. Private carriage may be conducted under 14 CFR part 91 or 125.

The term "for hire" is not defined in any of the FAA documents but is generally understood to mean that compensation for both direct and indirect expenses associated with the flight, as well as a profit margin for the operator, are collected from the person or persons benefiting from the flight operation.

The determination of whether the aircraft is large or small is based upon the definition provided in 14 CFR part 1. If the aircraft has maximum certificated takeoff weight of 12,500 pounds or more, it is a large aircraft. All aircraft less than 12,500 maximum certificated takeoff weight are considered to be small aircraft.

It may also help the reader understand when 14 CFR parts 121, 125, and 135 regulations apply, by taking a brief look at a list of flight operations where 14 CFR part 119 does not apply.

1. Student instruction
2. Nonstop sightseeing flights with less than 30 seats and less than 25 nautical miles (NM) from the departure airport
3. Ferry or training flights
4. Crop dusting or other agricultural operations
5. Banner towing
6. Aerial photography or surveying
7. Fire fighting
8. Powerline or pipeline patrol
9. Parachute operations on nonstop flights within 25 NM from the departure airport
10. Fractional ownership in accordance with 14 CFR part 91, subpart K

### *14 CFR Part 121—Operating Requirements: Domestic, Flag, and Supplemental Operations*

Title 14 CFR part 121 establishes the operational rules for air carriers flying for compensation or hire. A domestic operation is any scheduled operation (within the 48 contiguous states, the District of Columbia, or any territory or possession) conducted with either a turbo-jet aircraft, an airplane having 10 or more passenger seats, or a payload capacity greater than 7,500 pounds.

A "flag" operation means any scheduled operation (operating in Alaska or Hawaii to any point outside of those states, or to any territory or possession of the United States, or from any point outside the United States to any point outside the United States) conducted with either a turbo-jet aircraft, an airplane having 10 or more passenger seats, or a payload capacity greater than 7,500 pounds.

"Supplemental" operation means any common carriage operation conducted with airplanes having more than 30 passenger seats (if less than 30, the airplane must also be listed on the operations specifications of domestic and flag carriers), with a payload capacity of more than 7,500 pounds.

Part 121 operators are required by 14 CFR part 119 to have the following personnel:

- Director of Safety

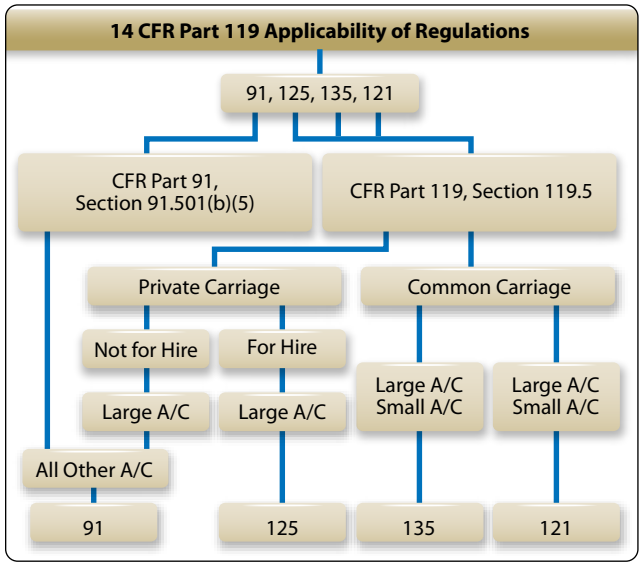



**Figure 2-8.** *Applicability of regulations.*

- Director of Operations
- Director of Maintenance
- Chief Pilot
- Chief Inspector

There are 28 subparts and 16 appendices in this regulation. However, only subparts J and L are of concern for the mechanic. Subpart J—Maintenance, Preventive Maintenance, and Alterations, identifies Special Airworthiness Requirements that deals with many of the mechanical aspects of a passenger or cargo aircraft. Subpart L—Maintenance, Preventive Maintenance, and Alterations, requires that a part 121 operator have an operational manual that contains the following information:

- Organizational chart
- List of individuals who may perform required inspections
- Company maintenance, preventive maintenance, or alterations
- A system to both preserve and retrieve maintenance and inspection related information

Also, 14 CFR part 121, section 121.1105, establishes the requirement for conducting inspections on aging aircraft.

### *14 CFR Part 125—Certification and Operations: Airplanes Having a Seating Capacity of 20 or More Passengers or a Maximum Payload Capacity of 6,000 Pounds or More; and Rules Governing Persons on Board Such Aircraft*

This regulation applies to private and noncommon carriage when such operations are conducted in airplanes having 20 or more seats (excluding crewmembers) or having a payload capacity of 6,000 pounds or more. There must also be "operations specifications" issued to the operator that include the following information:

- Kinds of operations authorized
- Types of aircraft and registration numbers of the airplanes authorized for use
- Approval of the provisions of the operator's manual relating to airplane inspections, together with the necessary conditions and limitations
- Registration numbers of the airplanes that are to be inspected under an approved airplane inspection program (AAIP) under 14 CFR part 125, section 125.247
- Procedures for the control of weight and balance of airplanes
- Any other item that the administrator determines is necessary

Just as in 14 CFR part 121, subpart E identifies special airworthiness requirements dealing mostly with the mechanical devices of the aircraft.

### *14 CFR Part 135—Operating Requirements: Commuter and On-Demand Operations and Rules Governing Persons on Board Such Aircraft*

As the title of this section states, this regulation is applicable to short distance commercial aircraft operations or "commuters" and nonscheduled carriers that operate "on-demand." These aircraft are frequently referred to as air taxi or air charter aircraft.

Aircraft operated under 14 CFR part 135 must be operated and maintained in accordance with the certificate holder's operations manual. This manual, when accepted by the FAA, specifies how the flight crew, ground personnel, and maintenance technicians conduct their operations.

A pivotal portion of this regulation is the first section in subpart J, 14 CFR part 135, section 135.411, Application. This section specifies that having a type certificated passenger seating configuration of nine or less may be maintained in accordance with the maintenance manual provided by the aircraft manufacturer. Those aircraft having a type certificated passenger seating configuration of 10 or more seats must be maintained in accordance with a maintenance manual written by the air carrier and must then be submitted to the FAA for approval. The requirements for the maintenance manual are specified in 14 CFR part 135, section 135.427. 14 CFR part 135, sections 135.415 through 135.417 and 135.423 through 135.443 specify additional maintenance requirements. 14 CFR part 135, sections 135.415 and 135.417 are applicable regardless of the number of seats in the aircraft.

A major change in the "nine or less" aircraft maintenance requirements occurred in February of 2005 when section 135.422, Aging Aircraft, was incorporated into 14 CFR part 135. This new subpart (note the even number) to 14 CFR 135 specifically prohibits a certificate holder from operating certain aircraft unless the Administrator has completed the aging aircraft inspection and records review. This inspection requires the certificate holder to show the FAA that the maintenance of age sensitive parts and components has been adequate to ensure safety.

This section only applies to multi-engine aircraft in scheduled operation with nine or fewer passenger seats. It does not apply to aircraft operating in Alaska. The required record review start date varies depending on the age of the aircraft.

However, once initiated, the repetitive inspection intervals are not to exceed 7 years.

The certificate holder must make both the aircraft and the records available to the FAA for inspection and review. The certificate holder must notify the Administrator at least 60 days in advance of the availability of the aircraft and the records for review.

The records must include the following information:

1. Total years in service of the airplane
2. Total time in service of the airframe
3. Date of the last inspection and records review required by this section
4. Current status of life-limited parts
5. Time since the last overhaul of all structural components required to be overhauled on a specific time basis
6. Current inspection status of the airplane, including the time since the last inspection required by the inspection program that the airplane is maintained under
7. Current status of applicable ADs, including the date and methods of compliance, and, if the AD involves recurring action, the time and date when the next action is required
8. A list of major structural alterations
9. A report of major structural repairs and the current inspection status of those repairs

### *14 CFR Part 145—Repair Stations*

This regulation underwent a major rewrite released in 2004 and was the most comprehensive change in nearly 20 years. It may be of interest to note an airframe and powerplant (A&P) certificate is not necessary to be employed at a repair station. The repair station may also employ both repairmen (under 14 CFR part 65, subpart E) and non FAA-certificated personnel. All work that is signed off is done so using the repair station certificate number and must be done only by persons authorized by 14 CFR part 65 to approve an article for return to service (RTS). Just as other certificate holders must have an operations manual, the repair station must have a repair station manual that contains the following:

- An organizational chart
- Procedures for maintaining rosters
- Description of housing, facilities, and equipment
- Procedures for revising the capability list and conducting a self-evaluation (audit)
- Procedures for revising the training program
- Procedures governing work done at another location
- Procedures for working on air carrier aircraft
- Description of the required records and record keeping
- Procedures for revising the repair station manual
- Description of the system to identify and control the sections of the manual

All records from repair station maintenance activity must be kept a minimum of 2 years. Domestic repair station certificates are effective until they are surrendered, suspended, or revoked. The certificates of foreign repair stations expire, usually after 1 or 2 years and must be renewed.

### *14 CFR Part 147—Aviation Maintenance Technician Schools*

Title 14 CFR part 147 defines the requirements for obtaining a maintenance training certificate. This certificate may be for either airframe, powerplant, or a combination of the two. The minimum number of curriculum hours for conducting either airframe or powerplant training independently is 1,150.

If both A&P ratings are offered, the combined total curriculum hours are 1,900. This is because of the 1,150 hours specified to obtain either the airframe or the powerplant rating, 400 hours are devoted to general studies. Only one set of general studies hours is applicable to the combined total. Therefore, 400 hours can be subtracted from the implied total of 2,300 hours (1,150 × 2) to obtain the reduced figure of 1,900 hours. Requirements are detailed as follows:

- Appendix A—Curriculum Requirements
- Appendix B—General Curriculum Subjects
- Appendix C—Airframe Curricular Subjects
- Appendix D—Powerplant Curriculum Subjects

### *14 CFR Part 183—Representatives of the Administrator*

As the aviation industry grows and the design, manufacture, and testing of aircraft gets more complex, the FAA faces both budget constraints and personnel shortages. As early as 1962, the FAA began a program to allow private sector persons in various areas of industry to be "designees" or "representatives of the FAA Administrator." These people are NOT FAA employees, but rather are designated by the FAA to act on their behalf. Regular doctors may serve as "aviation medical examiners," skilled pilots can become "pilot examiners," and experienced airframe and/or powerplant mechanics can become "designated mechanic examiners (DME)" to administer the oral and practical portion of the FAA testing.

Other lesser known designees are the designated engineering

representatives (DER), the designated manufacturing inspection representatives (DMIR), and the designated airworthiness representatives (DAR).

- DERs approve data based upon their engineering training and their knowledge of FAA regulations.
- DMIRs make conformity inspections only at their employer. They are similar to "designated repairmen" because they are only authorized to inspect parts at their employers' facility.
- DARs conduct aircraft certification and aircraft inspection functions on behalf of the FAA depending on specific functions they are authorized. They may perform work for either manufacturing facilities or maintenance entities depending on their designation.

## Explanation of Primary Regulations (Parts 43 and 91)

### 14 CFR Part 43—Maintenance, Preventative Maintenance Rebuilding, and Alteration

#### *Section 43.1—Applicability*

Paragraph (a) states quite clearly that aircraft (whether U.S.- or foreign-registered operating under 14 CFR part 121 or 135) and component parts thereof must be maintained in accordance with the rules set forth in this part. Although paragraph b states quite clearly the type of aircraft that this part does not apply to, it seems to have led to considerable confusion within the aviation industry. If an aircraft is flying with a Special Airworthiness—Experimental Certificate (FAA Form 8130-7, Special Airworthiness Certificate—pink color certificate) and that is the only airworthiness certificate this aircraft has ever had, then 14 CFR part 43 does not apply.

Conversely, sometimes during maintenance (especially STC modification—the STC is addressed later in this chapter), it becomes necessary to temporarily place the aircraft into Special Airworthiness—Experimental. This is done to show compliance with federal regulations. These aircraft must still be maintained in accordance with 14 CFR part 43, because the aircraft had a different kind of airworthiness (in this example a Standard) prior to being issued the Special Airworthiness Certificate.

#### *Section 43.2—Records of Overhaul and Rebuilding*

These terms are not defined in 14 CFR part 1 and are given full explanation in this subpart. Each term states that it may not be used to describe work done on an aircraft, airframe, aircraft engine, propeller, appliance, or component part unless that item has been:

- Disassembled
- Cleaned
- Inspected
- Repaired, as necessary
- Reassembled
- Tested

The key difference between the two terms is in determining how the item is tested. If it is "tested in accordance with approved standards acceptable to the Administration that have been developed and documented by the manufacturer, the item is said to be overhauled." This is basically another way of describing "service limits," a term frequently used to describe manufacturer specified acceptable limits for used parts. A "rebuilt item, on the other hand, must be tested to the same tolerances and limits as a new item."

#### *Section 43.3—Persons authorized to perform maintenance, preventive maintenance, rebuilding, and alterations*

There are nine different persons who may perform maintenance: (Reminder: Per 14 CFR part 1, the FAA definition of a person is "an individual, firm, partnership, corporation, association, joint-stock association, or governmental entity. It includes a trustee, receiver, assignee, or similar representative of any of them.")

1. Certificated mechanic, per 14 CFR part 65
2. Certificated repairman, per 14 CFR part 65
3. Person working under the supervision of a certificated mechanic or repairman
4. Holder of repair station certificate
5. Holder of an air carrier certificate
6. Except for holders of a sport pilot certificate, the holder of a pilot certificate issued under part 61 may perform preventive maintenance on any aircraft owned or operated by that pilot which is not used under 14 CFR part 121, 129, or 135. The holder of a sport pilot certificate may perform preventive maintenance on an aircraft owned or operated by that pilot and issued a special airworthiness certificate in the light-sport category.
7. Pilot of a helicopter (when operated under 14 CFR part 135 and in remote areas) may perform specific preventive maintenance actions. These actions may only be accomplished under the following conditions:
   - The mechanical difficulty or malfunction occurred en route to or in the remote area.
   - The pilot has been satisfactorily trained and is

authorized in writing by the certificate holder to perform the required maintenance.

- There is no certificated mechanic available.
- The certificate holder has procedures to evaluate the work performed when a decision for airworthiness is required. The work done is listed in paragraph (c) of Appendix A of this chapter.

8. Holder of part 135 certificate may allow pilots of aircraft with nine or less passenger seats to remove and reinstall cabin seats and stretchers and cabin mounted medical oxygen bottles. These actions may only be accomplished under the following conditions:
   - The pilot has been satisfactorily trained and is authorized in writing by the certificate holder to perform the required maintenance.
   - The certificate holder has written procedures available to the pilot to evaluate the work performed.
9. Manufacturer may inspect and rebuild any item it has manufactured.

### *Section 43.5—Approval for return to service after maintenance, preventive maintenance, rebuilding, and alterations*

Approving an aircraft component for return to service after maintenance, preventive maintenance, rebuilding, or alteration must be done by creating an appropriate maintenance record entry as required by either 14 CFR part 43, section 43.9 or 43.11. This may include the use of FAA Form 337, Major Repair and Alteration, if the maintenance action was a major repair or a major alteration. Whenever a maintenance action is being planned, it is critical that the technician understands exactly:

1. What he/she is going to do.
2. How that work is classified by the FAA.
3. What type of documentation is required to support this activity.

First consider whether this a repair or an alteration. This should be a relatively simply decision since a repair basically returns the aircraft to its previous or unaltered condition (i.e., replacing magnetos, an exhaust system, tires, or brakes). Even replacing an entire engine (although it is a big job) is still a repair if it is the one properly specified for that aircraft. An alteration on the other hand, always changes or modifies the aircraft from its previous state (i.e., installing winglets, new avionics, or an engine that is not listed in the aircraft TCDS).

The second question to consider is whether or not the work that to be performed constitutes a major or a minor maintenance action. A "major" action is typically one that might appreciably affect weight, balance, structural strength, performance, powerplant operation, flight characteristics, or other qualities affecting airworthiness and that are not done according to accepted practices or cannot be done by elementary operations. This is a much more complex question, but it is extremely important as it drives the final question concerning the substantiating documentation. Please refer to 14 CFR part 1 and part 43, appendix A, for additional clarification and examples.

The third question deals with the type of documentation required to substantiate the work performed. Minor repairs and alterations need only to refer to "acceptable" data, such as manufacturers' maintenance manuals or AC 43.13-1. The maintenance action can simply be recorded in the maintenance record as a logbook entry. Major repairs and alterations require "approved data." Some examples of approved data are AD notes, STCs, TCDS, DER-specific delegations, and FAA-approved manufacturer Service Bulletins (SB).

Sometimes the repair or alteration being performed does not have previously-approved data. In that case, the technician may request that the FAA accomplish a "Field Approval." In this procedure, the technician completes the front side of Form 337 through block 6 (leaving block 3 open for later FAA approval) and then indicates in block 8 on the back what work is to be done and what the substantiating reference data is. Form 337 is then submitted to the local FAA FSDO office for review and approval by an ASI. If necessary, this ASI may seek input from other ASIs or FAA specialists to assist in the review of the data. If the data is found to comply with FAA regulations, the ASI enters one of the following statements in block 3, depending on whether the ASI has performed a review of the data only or has physically inspected the aircraft:

- "The technical data identified herein has been found to comply with applicable airworthiness requirements and is hereby approved for use only on the above described aircraft, subject to conformity inspection by a person authorized in 14 CFR part 43, section 43.7."

  or

- "The alteration or repair identified herein complies with the applicable airworthiness requirements and is approved for use only on the above described aircraft, subject to conformity inspection by a person authorized in 14 CFR part 43, section 43.7."

### *Section 43.7—Persons authorized to approve aircraft, airframes, aircraft engines, propellers, appliances, or component parts for return to service after maintenance, preventive maintenance, rebuilding, or alteration*

There are seven different persons listed in this section who may sign RTS documentation:

1. Certificated mechanic or holder of an inspection authorization (IA).
2. Holder of a repair station certificate.
3. Manufacturer.
4. Holder of an air carrier certificate.
5. Certificated private pilot.
6. Repairman certificated with a maintenance rating for light sport aircraft (LSA) only.
7. Certificated sport pilot for preventive maintenance on an aircraft owned and or operated by them.

Note that although a certificated repairman is authorized to work on a product undergoing maintenance, preventive maintenance, rebuilding, or alterations (refer to 14 CFR part 43, section 43.3), they are not authorized to approve that product for RTS. They must make the appropriate maintenance record entry per the requirements of 14 CFR part 43, section 43.9 or 43.11.

### *Section 43.9—Content, form and disposition of maintenance, preventive maintenance, rebuilding, and alteration records (except inspection performed in accordance with parts 91 and 125, and sections 135.411(a)(1) and 135.419 of this chapter)*

The first observation is that this section specifically excludes inspection entries (those are covered in 14 CFR part 43, section 43.11). This section deals exclusively with maintenance record entries.

The next observation is that the list of maintenance actions includes "preventive maintenance." As stated in the explanation of 14 CFR part 43, section 43.3, a certificated pilot is authorized to perform preventive maintenance on the aircraft they own or operate. Therefore, remember that the pilot must make a record entry of the preventive maintenance they have accomplished. There are three distinct issues to be addressed in the maintenance entry and they answer the questions of "what?, when?, and who?"

- What—a description of the work performed
- When—the date the work was completed
- Who—the name of the person who did the work if other than the person who approves the RTS
  —the signature, certificate number, and type of certificate of the person who is approving the work for RTS

***Note:*** Frequently, logbooks have a statement entered that ends something like this: " … and is hereby returned to service. Joe Fixer A&P, Certificate #123456789." As this section of the regulation currently reads, that part of the record entry is not required. Title 14 of the CFR part 43, section 43.9 clearly states that "the signature constitutes the approval for return to service only for the work performed." Furthermore, the technician is only signing off the work they have done. Later, 14 CFR part 43, section 43.11 explains that an inspection write-up usually carries a broader scope of responsibility. This section is very clear that the entry completed in accordance with this section only holds the technician responsible for the service maintenance action they entered.

If the maintenance accomplished was a major repair or alteration, the work must be documented on FAA Form 337 and requires supporting approved data. If the maintenance action was a major repair and it was done by a certificated repair station, a signed copy of the completed customer work order accompanied by a signed maintenance release may be used in lieu of the FAA Form 337.

### *Section 43.10—Disposition of Life-Limited Aircraft Parts*

(Note the even number again. This regulation became part of 14 CFR part 43 in 2002.)

This section presents two terms not previously defined in 14 CFR:

1. Life-limited part means any part that has specified a mandatory replacement limit.
2. Life status means the accumulated cycles, hours, or any other mandatory limit of a life-limited part.

This section then goes on to specify what to do with parts that are temporarily removed from and then reinstalled on a type-certificated product; what to do with parts that are removed from a type certified product and not immediately reinstalled; and how to transfer life-limited parts from one type-certificated product to another.

When a life-limited part is removed, the person removing it from the type-certificated product must control the part and ensure proper tracking of the life-limiting factor. This is to prevent the installation of the part after it has reached its life limit. There are seven possible methods the technician or repair facility may choose from to comply with this requirement.

1. Recordkeeping
2. Tagging
3. Non-permanent marking
4. Permanent marking
5. Segregation
6. Mutilation
7. Any other method approved or accepted by the FAA

When a life-limited part is transferred, the information concerning the life status of that part must be transferred with it. Although regulations already did exist that required the tracking of life-limited parts when they were installed on an aircraft, this regulation was generated to govern the disposition of such parts when they were removed from the aircraft.

### *Section 43.11—Content, form, and disposition of records for inspections conducted under parts 91 and 125, and sections 135.411(a)(1) and 135.419 of this chapter*

This section deals exclusively with inspection record entries; however, the requirements are similar to 14 CFR part 43, section 43.9 in that information of what, when, and who is required.

- What—type of inspection, including a brief description
- When—date of the inspection and the total time in service
- Who—the signature, certificate number, and kind of certificate of the person approving or disapproving the RTS

Since this is an inspection write-up and not a maintenance entry, it is quite possible that the inspecting technician could reject or disapprove the item being inspected for the RTS. When that situation occurs, the regulation states in paragraph (b) that a list of discrepancies must be given to the owner. A reference to this list and its delivery to the aircraft owner must be reflected in the record entry. Although the regulation neither specifies how those discrepancies can be cleared, nor who may do them, any appropriately-rated repair station or certificated technician can perform the required maintenance actions. When they are completed and the proper maintenance record entries are generated in accordance with 14 CFR part 43, section 43.9, the aircraft is approved for RTS. It is neither necessary to have an additional inspection, nor is it necessary to contact the disapproving inspector.

If the aircraft is on a progressive inspection program, the inspection statement changes slightly from the statement referenced earlier by adding the reference to both a "routine inspection" and a "detailed inspection." Refer to explanatory text of 14 CFR part 43, section 43.15 for a definition of these terms. Inspections accomplished in accordance with other inspection program requirements must identify that particular program and that part of the program the inspection completed.

### *Section 43.12—Maintenance Records: Falsification, Reproduction, or Alteration*

The aviation community relies heavily on trust and honesty in both oral and written communication. The maintenance log entries described in 14 CFR part 43, sections 43.9 and 43.11 provide the documentation trail relied upon by aircraft owners, pilots, and technicians regarding the aircraft's maintenance history. Falsification of these records is potentially dangerous to the personnel who rely on the accuracy of these records.

This section identifies that fraudulent entries are unacceptable. If someone commits such an act, that action is the basis for suspension or revocation of the appropriate certificate, authorization, or approval. A technician who is encouraged by their employer, or by anyone else, to falsify records in any way should remember this comment: "Companies come and go, but my signature lasts a lifetime. I will not use it inappropriately."

### *Section 43.13—Performance Rules (General)*

This section deals with the specific requirements for conducting maintenance. (***Note:*** This section best reflects the relationship between the FAA's numbering of ACs and the regulations they are related to.) Paragraph 3 on the cover page of AC 43.13-2B, Acceptable Methods, Techniques, and Practices—Aircraft Alterations, dated March 3, 2008 states:

"Title 14 of the Code of Federal Regulations (14 CFR) part 43, section 43.13(a) states that each person performing maintenance, alteration, or preventive maintenance on an aircraft, engine, propeller, or appliance must use the methods, techniques, and practices prescribed in the current manufacturer's maintenance manual or Instructions for Continued Airworthiness prepared by its manufacturer, or other methods techniques or practices acceptable to the Administrator, except as noted in section 43.16." *[Figure 2-9]*

Although not all ACs are linked this directly, there is a definite relationship between ACs and companion regulations. Refer to this chapter on ACs for additional information.

Aircraft maintenance technicians (AMTs) are highly skilled personnel, because aviation maintenance work requires great attention to detail. The complexity of technology on today's aircraft demands a significant level of communication to

properly accomplish maintenance, preventive maintenance, rebuilding, or alteration. This communication frequently comes in written form (i.e., manufacturer's maintenance manuals or ICA). If neither of these documents provide the guidance the technician needs to perform maintenance, either AC 43.13 (AC 43.13-1 or AC 43.13-2) contain examples of "other methods, techniques, or practices acceptable to the Administrator" that may be sufficient. However, these ACs specifically state that the information is applicable to non-pressurized areas of civil aircraft weighing 12,500 lb gross weight or less.

In addition to the documentation, the technician must also use the proper tools, equipment, and test apparatus that ensures that the work complies with accepted industry practices. If the test equipment specified by the manufacturer is not available, equipment that is determined to be equivalent and acceptable to the Administrator may be used. The technician should be cautious, however, as "proving" the equivalence of test equipment may not be as simple as it seems.

Air carriers (commercial—"scheduled" airlines operating under 14 CFR part 121, the "commuter/on demand" aircraft operating under 14 CFR part 135, and foreign air carriers and operators of U.S.-registered aircraft under 14 CFR part 129) may use the maintenance manual required by the operations specifications to comply with the requirements of this section. The operator must provide a continuous airworthiness maintenance and inspection program acceptable to the Administrator.

### *Section 43.15—Additional Performance Rules for Inspections*

This section presents general comments concerning the responsibility of conducting an inspection and then provides details of three separate conditions. They are rotorcraft, annual and 100-hour inspections, and progressive inspections.

1. Rotorcraft—If a rotorcraft is being inspected, specific items, such as rotor transmissions and drive shafts, must be inspected.
2. Annual and 100-hour inspections—When performing an annual or 100-hour inspection, a checklist must be used. This checklist may be a personal one or one from the manufacturer. Either way, it must include the scope and detail of the inspection in Appendix D. Specific engine performance is also required to be tested (or monitored) as part of RTS for an annual or 100-hour inspection. This applies whether the aircraft is reciprocating or turbine powered.
3. Progressive inspection—If a progressive inspection is being conducted, it must be preceded by a complete aircraft inspection. (***Note:*** A progressive inspection is the result of breaking down the large task of conducting a major inspection into smaller tasks that can be accomplished periodically without taking the aircraft out of service for an extended period of time.) Two new definitions are also presented: "routine" and "detailed." A routine inspection is a visual examination or check of the item, but no disassembly is required. A detailed inspection is a thorough examination of the item, including disassembly. The overhaul of a component is considered to be a detailed inspection.

   If the aircraft is away from the station where inspections are normally conducted, an appropriately rated mechanic, a certificated repair station, or the manufacturer of the aircraft may perform inspections in accordance with the procedures and using the forms of the person who would otherwise perform the inspection.

### *Section 43.16—Airworthiness Limitations*

The technician performing inspection or maintenance actions on an aircraft must be certain they have all appropriate data available. Each person performing an inspection or other maintenance specified in an Airworthiness Limitations section of a manufacturer's maintenance manual or Instructions for Continued Airworthiness shall perform the inspection or other maintenance in accordance with that section, or in accordance with operations specifications approved by the Administrator under part 121 or 135, or an inspection program approved under 14 CFR part 91, section 91.409(e). ICAs, as required by 14 CFR part 21, section 21.50, must also be consulted when available. Since 1998, the FAA has required ICAs to be generated for all major alterations that are accomplished by the field approval process. This section specifies that the technician is responsible to perform inspections or maintenance specified in an airworthiness limitation in accordance with all the preceding instructions.

### *Section 43.17—Maintenance, preventive maintenance, or alterations performed on U.S. aeronautical products by certain Canadian persons*

This section was significantly revised in 2005, as the result of a Bilateral Aviation Safety Agreement (BASA) between the United States and Canada. This section of 14 CFR part 43 defines some terms and gives specific limitations as to what an Aircraft Maintenance Engineer (AME is the Canadian equivalent to the U.S. A&P) may do to maintain U.S.-registered aircraft located in Canada. It also provides similar limitations for an Approved Maintenance Organization. (AMO is the Canadian equivalent to the U.S.-certified repair stations.)

### *Appendix A—Major Alterations, Major Repairs, and*

U.S. Department
of Transportation
Federal Aviation
Administration

# Advisory Circular

**Subject:** Acceptable Methods, Techniques, and Practices – Aircraft Alterations

**Date:** 3/3/08

**AC No:** 43.13-2B

**Initiated by:** AFS-300

**1. PURPOSE.** This advisory circular (AC) contains methods, techniques, and practices acceptable to the Administrator for the inspection and alteration on non-pressurized areas of civil aircraft of 12,500 lbs gross weight or less. This AC is for use by mechanics, repair stations, and other certificated entities. This data generally pertains to minor alterations; however, the alteration data herein may be used as approved data for major alterations when the AC chapter, page, and paragraph are listed in block 8 of FAA Form 337 when the user has determined that it is:

**a.** Appropriate to the product being altered,

**b.** Directly applicable to the alteration being made, and

**c.** Not contrary to manufacturer's data.

**2. CANCELLATION.** AC 43.13-2A, Acceptable Methods, Techniques, and Practices— Aircraft Alterations, dated January 1, 1977, is canceled.

**3. REFERENCE.** Title 14 of the Code of Federal Regulations (14 CFR) part 43, § 43.13(a) states that each person performing maintenance, alteration, or preventive maintenance on an aircraft, engine, propeller, or appliance must use the methods, techniques, and practices prescribed in the current manufacturer's maintenance manual or Instructions for Continued Airworthiness prepared by its manufacturer, or other methods, techniques, or practices acceptable to the Administrator, except as noted in § 43.16. FAA inspectors are prepared to answer questions that may arise in this regard. Persons engaged in the inspection and alteration of civil aircraft should be familiar with 14 CFR part 43, Maintenance, Preventive Maintenance, Rebuilding, and Alterations, and part 65, subparts A, D, and E of Certification: Airmen Other than Flight Crewmembers, and applicable airworthiness requirements under which the aircraft was type-certificated.

**4. COMMENTS INVITED.** Comments regarding this AC should be directed to DOT/FAA: ATTN: Aircraft Maintenance Division, 800 Independence Ave., SW., Washington, DC 20591, FAX (202) 267-5115.

ORIGINAL SIGNED By

James J. Ballough
Director Flight Standards Service

**Figure 2-9.** *AC 43.13-2B Excerpt.*

### *Preventive Maintenance*

This appendix provides a comprehensive, but not exclusive, list of subjects. For instance, paragraph (a) is titled Major Alteration, and is further subdivided as follows:

- Airframe
- Powerplant
- Propeller
- Appliance

This same subdivision is used in paragraph (b), Major Repairs. Paragraph (c), Preventive Maintenance, identifies those maintenance actions that are defined as preventive maintenance, provided the maintenance does not involve complex assembly operations. Preventive maintenance work may be accomplished by the holder of at least a private pilot certificate provided they are the owner or operator of that aircraft, and it is not operated under 14 CFR part 121, 129, or 135.

### *Appendix B—Recording of Major Repairs and Major Alterations*

In most cases when a major repair or alteration is accomplished, FAA Form 337, Major Repair or Alteration, is completed at least in duplicate with the original going to the aircraft owner and a copy sent to the FAA Aircraft Registration Branch in Oklahoma City where all civil aircraft information is compiled and retained. ***Note:*** Historically, the second copy was sent to the local FAA FSDO within 48 hours after RTS. This copy is reviewed by an ASI and then forwarded by the FSDO to FAA records in Oklahoma City. However, in the fall of 2005, the FAA made a significant change to this submittal process and now requires the technician to submit the Form 337 directly to the Aircraft Registration Branch in Oklahoma City. Although a third copy is not required, it makes good business sense for the technician or certified repair station to keep a copy of the work that was accomplished.

However, if a certificated (part 145) repair station completes a major repair, it may provide the customer with a signed copy of the work order and a maintenance release signed by an authorized representative of the repair station, instead of the FAA Form 337. If the major repair or alteration was done by an AME or AMO, the copy normally provided to the FAA-FSDO is sent directly to the FAA Aircraft Registration Branch.

However, if extended range tanks are installed in either passenger or cargo compartments, the technician must generate a third FAA Form 337 for the modification. This copy must be placed and retained in the aircraft. (Refer to 14 CFR part 91, section 91.417(d).)

### *Appendix C—(Reserved)*

Appendix C is reserved for future use and therefore currently contains no information.

### *Appendix D—Scope and Detail of Items To Be Included in Annual and 100-Hour Inspections*

Some important items to consider in this appendix are:

1. The list of items and areas to be inspected are exactly the same for an annual as a 100-hour inspection. The difference between the inspections is in who is authorized to perform and approve the aircraft for RTS following the inspection. Refer to 14 CFR part 65, section 65.95(a)(2) that states that an IA must perform an annual inspection.
2. The aircraft and engine must be cleaned prior to conducting the inspection.
3. Any miscellaneous item not covered in the detailed list provided must also be inspected for improper installation and operation.
4. There are eight specific areas identified for detailed inspection. They are the fuselage hull group, cabin/flight deck group, engine/nacelle group, landing gear group, wing/center section group, empennage assembly, propeller group, and the radio group.

### *Appendix E—Altimeter System Test and Inspection*

This is commonly referred to as "the 411 test." Refer to 14 CFR part 91, section 91.411 that requires that no person may operate an aircraft in controlled airspace under IFR unless the aircraft has had this test completed successfully within the preceding 24 months.) This section requires detailed testing of the static pressure system, the altimeter, and the automatic pressure altitude reporting equipment, and that the test information be recorded in the maintenance logs and on the altimeter.

### *Appendix F—ATC Transponder Tests and Inspections*

This is commonly referred to as "the 413 test." (Refer to 14 CFR part 91, section 91.413, which requires that no person may use a transponder unless it has had this test completed successfully within the preceding 24 months.)

This section specifies complex sets of tests, which may be accomplished either as a bench test or by using portable test equipment. Major categories of the testing required are radio reply frequency, suppression, receiver sensitivity, radio frequency peak output power, and mode S (when applicable). Upon completion of testing, proper entries must be made in the maintenance record.

## 14 CFR Part 91—General Operating and Flight Rules

### *Subpart A—General*

As mentioned in the brief overview of the regulation portion earlier in this chapter, this part is actually addressing the operation of the aircraft. For example, 14 CFR part 91, section 91.7(a) states "no person may operate a civil aircraft unless it is in an airworthy condition." We learned earlier that this term means that the aircraft conforms to its approved type design and is in condition for safe operation. When the pilot performs a preflight inspection, they are making a determination concerning the "condition for safe operation." The pilot does not usually determine "conformity to type design" unless they perform a review of the maintenance records. However, since that is fundamental to the definition of airworthy, it is still part of their responsibility. Therefore, a professional and ethical technician wants to help the customer understand their responsibilities in maintaining and documenting the airworthiness of the aircraft.

### *Subpart E—Maintenance, Preventive Maintenance, and Alterations*

*Section 91.401—Applicability*

Although this subpart describes in general the rules regarding maintenance, preventive maintenance, and alteration, certain sections do not apply if the aircraft is operated in accordance with 14 CFR part 121, 125, 129, or 135.

*Section 91.403—General*

The owner/operator holds the primary responsibility for maintaining the aircraft in airworthy condition. This includes compliance with all applicable ADs and is the reason that the FAA sends new AD notes to the registered owners of the affected aircraft. All maintenance performed must be accomplished in accordance with 14 CFR part 43. Compliance with the appropriate manufacturer maintenance manuals and ICA is also required. Mandatory replacement times, inspection intervals, and related procedures as outlined in the FAA-approved operations specifications must also be complied with.

*Section 91.405—Maintenance Required*

The owner/operator is required to have the appropriate inspections made, and to have discrepancies repaired in accordance with part 43. They are also required to ensure that the appropriate entries have been made in the maintenance records. Any inoperative instruments or equipment must be properly placarded as inoperative.

*Section 91.407—Operation after maintenance, preventive maintenance, or alteration*

Whenever the aircraft has undergone maintenance, preventive maintenance, rebuilding or alteration, it must have been approved for RTS and a proper entry made in the maintenance records. If the maintenance that was done could have appreciably changed the flight characteristics, an appropriately rated pilot must perform an operational flight check of the aircraft and must make an entry of the flight in the maintenance records. If ground testing and inspection can show conclusively that the maintenance has not adversely affected the flight characteristics, no flight test is required.

*Section 91.409—Inspections*

This paragraph identifies various types of inspection applicable to the civilian aircraft fleet. Paragraph (a) defines the requirement for an annual inspection. However, there are certain exceptions to this regulation:

1. An aircraft that carries a special flight permit, a current experimental certificate, or a light-sport or provisional airworthiness certificate;
2. An aircraft inspected in accordance with an approved aircraft inspection program under part 125 or 135 of this chapter and so identified by the registration number in the operations specifications of the certificate holder having the approved inspection program;
3. An aircraft subject to the requirements of paragraph (d) or (e) of this section; or
4. Turbine-powered rotorcraft when the operator elects to inspect that rotorcraft in accordance with paragraph (e) of this section.

Annual inspections are usually the inspection method associated with small "general aviation" aircraft. If this same aircraft is used for hire (including flight instruction for hire), then the aircraft must also be inspected every 100 hours of time in service. This requirement for a 100-hour inspection to be conducted on an aircraft may be exceeded by as much as 10 hours if the aircraft is en route to reach a facility that will be conducting the inspection. Any time accrued between 100 and 110 hours is subtracted from the hours remaining before the next 100-hour inspection.

Since aircraft used for hire only generate revenue when they are flying, any time that the aircraft is "down for inspection" can result in a loss of income for the owner/operator. Therefore, the FAA has made provision to minimize the impact of the 100-hour and annual inspection requirement. The owner/operator may petition the local FSDO for approval of a progressive inspection program. This program breaks the complete inspection of the aircraft into smaller, less time-consuming steps. (Refer to 14 CFR part 43, Appendix D.) This inspection may be either performed or supervised by a technician holding an IA. The program must ensure at all times that the aircraft is airworthy. The owner/operator must submit an inspection schedule with their application

to the FAA. This schedule must identify the time intervals (hours or days) when routine and detailed inspections are to be accomplished. (Refer to 14 CFR part 43, section 43.15.) Just as with the 100-hour inspection, a 10-hour maximum extension of a specified inspection interval is allowed if the aircraft is en route. A change in the inspection interval is also allowed for changes in service experience. If the progressive inspection is discontinued, the aircraft is again subject to the traditional annual and 100-hour inspections.

Other inspection programs that may be applicable to other aircraft are a continuous airworthiness inspection program and an approved aircraft inspection program (AAIP). The former program is applicable to either a part 121 or 135 carrier, but the latter program is limited to part 135 operators only. Finally, the owner/operator may use either a current inspection program recommended by the aircraft manufacturer or one established by the owner/operator and approved by the local FSDO. Any subsequent changes to that program must also be approved by the local FSDO.

There may be an instance when the operator of an aircraft wishes to change from one type of inspection program to another. In that case, the time in service, calendar times, or cycles of operation from the current program must be carried over to the subsequent program.

*Section 91.411—Altimeter System and Altitude Reporting Equipment Tests and Inspections*

Commonly referred to as "the 411 test," this section specifies the requirements for testing the static pressure system, each altimeter instrument, and each automatic pressure altitude reporting system every 24 calendar months. The static system must also be tested any time it has been "opened and closed," except for the normal use of the system drain and alternate static system pressure valves. If the automatic pressure altitude reporting system of the air traffic control (ATC) transponder is either installed or subjected to maintenance actions, the system must also be tested per Appendix E of 14 CFR part 43.

Due to the inherent design and accuracy of this system, only the aircraft manufacturer, a properly-rated repair station, or a certificated airframe mechanic may perform these tests. The airframe technician may only perform the inspection and test of the static pressure system. Calibration and maintenance of related instruments is specifically prohibited to the technician by the language of 14 CFR part 65, section 65.81 and specifically allowed in 14 CFR part 145, section 145.59 for repair stations holding an instrument rating.

TSO'd items are considered to be "tested and inspected" as of the date they were manufactured. The maximum altitude that the system was tested is the maximum altitude that the aircraft can be flown instrument flight rules (IFR) in controlled airspace.

*Section 91.413—ATC Transponder Tests and Inspections*

This "413 test" is the other test required every 24 months. Whenever the ATC transponder is installed or has undergone maintenance, the complete system must be tested and inspected in accordance with Appendix E of 14 CFR part 43. The transponder itself must be tested and inspected in accordance with Appendix F of 14 CFR part 43. As with the 411 test, only certain persons are authorized to conduct the tests. They are the manufacturer of the aircraft, a properly certificated repair station, or the holder of a continuous airworthiness maintenance program under 14 CFR part 121 or 135.

*Section 91.415—Changes to Aircraft Inspection Programs*

If the FAA determines that the inspection program established and approved under either 14 CFR part 91, section 91.409 or 91.1109 must be revised to ensure continued safety and adequacy of the program, the owner/operator must make the necessary changes as identified by the Administrator. If the owner/operator desires to contest this request, they must petition the FAA to reconsider their request to change the program within 30 days of receiving the change request from the FAA.

*Section 91.417—Maintenance Records*

The understanding and implementation of this section is fundamental to the aircraft industry, in general, and the aircraft owner/operator, in specific. A professional maintenance technician must be knowledgeable of this section and be able to help the owner/operator understand it. *[Figure 2-10]* This section identifies four types of records—two are quite specific (paragraphs a and d) and two are more general: (a)(1) and (a)(2). Paragraph (a) refers to the 411 and 413 testing that requires testing every 24 months. Therefore, records must be kept for that length of time. Paragraph (d) refers to the installation of fuel tanks in the cabin or cargo area. The FAA Form 337 authorizing this installation must be kept on board the aircraft all the time.

***Note:*** Other than this paragraph, there is no requirement that the maintenance records of the aircraft be carried on the aircraft. In fact, there are very logical reasons to not do so in most cases. The two biggest concerns are damaged or lost records. It is much safer to retain the logs in a filing system in the office. It is also a very wise idea to have the logbook copied or scanned and retained at a separate location should a catastrophic event (fire, flood, tornado, hurricane, and so forth) occur at the site the original records are retained.

Subparagraph (a)(1) then lists those records that are later

defined in (b)(1) as being retained for 1 year or until the work is repeated or superseded. Subparagraph (a)(2) specifies the records that are permanent records and are identified in subparagraph (b)(2) as those that must be transferred with the aircraft. Refer to the chart for further clarification. *[Figure 2-10]*

Paragraph (c) requires that all of the maintenance records mandated by this section be made available upon request to the Administrator or any authorized representative of the NTSB. Furthermore, the owner/operator must provide the Form 337 required to be aboard the aircraft whenever additional fuel tanks are installed in either the passenger compartment or the baggage compartment, per paragraph (d), to any law enforcement officer upon request.

*Section 91.419—Transfer of Maintenance Records*

When an aircraft is sold, it is logical that the records are transferred with it. They may be either in plain language or coded. The purchaser may elect to permit the seller to retain the actual records; however, if that occurs the purchaser (now the current owner/operator) must still make these records available to either the FAA or the NTSB upon request.

*Section 91.421—Rebuilt Engine Maintenance Records*

This section presents the term "zero time." Although not truly given as a definition, the wording of the regulation is very clear that an aircraft engine, when rebuilt by the engine manufacturer or an agency approved by the manufacturer, may be given a new maintenance record showing no previous operating history. This new record must include a signed statement with the date it was rebuilt, any changes incorporated by compliance with AD notes, and compliance with any of the manufacturer's SB.

## Civil Air Regulations (CAR)

Prior to 1926, access to flying was uncontrolled. No licensing or certification was required. By the middle of the 1920s, it became obvious that unregulated private and commercial flying was dangerous. There was a growing awareness and acceptance that regulation could improve safety and encourage growth in aviation. Therefore in 1926, the aviation industry requested Congress to enact federal legislation to regulate civil aviation. Thus, the Air Commerce Act of 1926 provided for the:

1. Establishment of airways.
2. Development of aviation aids.
3. Investigation of aviation accidents.
4. Licensing of pilots.
5. Certification of aircraft.

The Civil Air Regulations (CARs) were part of the original certification basis for aircraft first certified in the 1940s, 1950s, and 1960s by the Civil Aviation Authority (CAA). Therefore, the CARs may still be needed as a reference for these older aircraft or as a standard for minor changes to older aircraft designs. *[Figure 2-11]*

### CAR 3—Airplane Airworthiness—Normal, Utility, Aerobatic, and Restricted Purpose Categories

As the name implies, this specific regulation is the basis for the current 14 CFR part 23 regulation [*Figure 2-1*]. It has the following subpart categories:

- A—Airworthiness Requirements
- B—Flight Requirements—General
- C—Strength Requirements—General
- D—Design and Construction—General
- E—Powerplant Installations—Reciprocating Engines
- F—Equipment

Some examples of CAR 3 aircraft are Piper PA 22, PA 28, PA 32, and Cessna 182, 195, and 310.

***Note:*** The "CAR" acronym actually has two interpretations: Civil Air Regulations and Canadian Aviation Regulations. The technician must clearly understand the difference and recognize when one or the other is appropriate.

### CAR 4a—Airplane Airworthiness

This regulation was originated in 1936 and last amended on December 15, 1952. The subparts included in this regulation are:

- A—Airworthiness Requirements
- B—Definitions
- C—Structural Loading Conditions, General Structural Requirements
- D—Proof of Structure
- E—Detail Design and Construction
- F—Equipment
- G—Powerplant Installation
- H—Performance
- I—Miscellaneous Requirements

Initially, this regulation was the basis for establishing the design requirements for virtually all produced aircraft in the 1930s, 1940s, and 1950s. Eventually CAR 3 evolved as the regulatory material specific to small aircraft, and CAR 4a and b focused on regulatory requirements for large aircraft.

It is very important to review the TCDS for each aircraft. For example, The Cessna 140 was certified as a landplane under CAR 3, but under CAR 4a as a ski-plane or seaplane. Another example of a more current and larger aircraft is the Gulfstream 1159 and 1159A. The former is certified under CAR 4b, but the latter is certified to 14 CFR part 25.

## Suspected Unapproved Parts (SUP)

There are four types of aircraft parts:

1. Good parts with good paperwork.
2. Good parts with bad paperwork.
3. Bad parts with "good" (bogus) paperwork.
4. Bad parts with bad paperwork.

The first of those listed represents properly authorized parts that, when properly installed, are approved parts, and the aircraft can be returned to service. The last of those listed represent unauthorized and unapproved parts. The technician should be alert for these and must never install them on an aircraft.

The center two categories of parts represent suspected unapproved parts. If either the physical part or the paperwork associated with the part is questionable, it is best to contact the shop foreman, shift supervisor, or the assigned quality individual to discuss your concerns. Suspected unapproved parts (SUPs) should be segregated and quarantined until proper disposition can be determined. Contacting the manufacturer of the product is a good way to start gathering the facts concerning the product in question. Refer to the current version of AC 21-29, Detecting and Reporting Suspected Unapproved Parts, for additional information. Current contact information for submitting a SUP Notification can be found at www.faa.gov.

## Other FAA Documents

### Advisory Circulars (AC)

AC refers to a type of publication offered by the FAA to provide guidance for compliance with airworthiness regulations. They provide guidance such as methods, procedures, and practices acceptable to the Administrator for complying with regulations. ACs may also contain explanations of regulations, other guidance material, best practices, or information useful to the aviation community. They do not create or change a regulatory requirement. The AC system became effective in 1962. It provides a single, uniform, agency-wide system that the FAA uses to deliver advisory material to FAA customers, industry, the aviation community, and the public.

Unless incorporated into a regulation by reference, the content of ACs are not binding on the public. ACs are issued in a numbered-subject system corresponding to the subject areas of the FARs (14 CFR, Chapter 1, Federal Aviation Administration) and Chapter 3, Commercial Space Transportation, Federal Aviation Administration, Department of Transportation, Parts 400–450. An AC is issued to provide guidance and information in a designated subject area or to show a method acceptable to the Administrator for complying with a related federal aviation regulation.

Because of their close relationship to the regulations, ACs are arranged in a numbered system that corresponds to the subject areas of the CFRs. In some series, consecutive numbers may

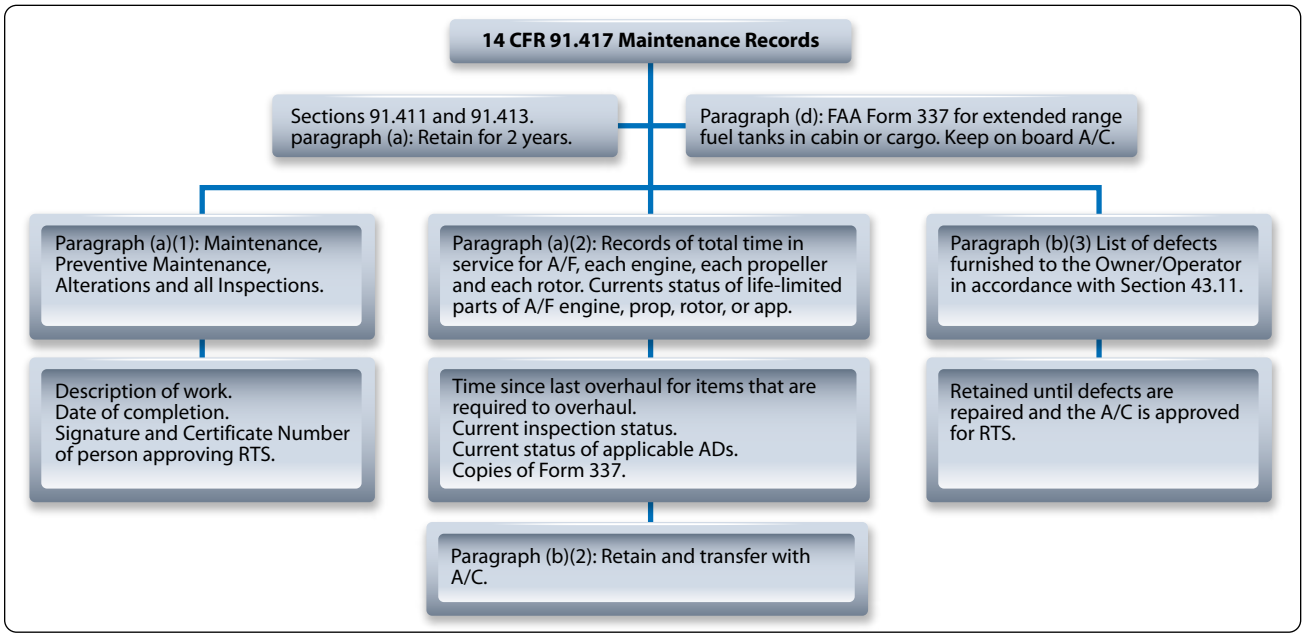



**Figure 2-10.** *Maintenance records.*

| Predecessor Regulations to the Federal Aviation Regulations (14 CFR) | |
|---|---|
| **Aeronautical Bulletins** | |
| 7A | Airworthiness Requirements for Aircraft |
| 7F | Airworthiness Requirements for Aircraft Components and Accessories |
| 7G | Airworthiness Requirements for Engines and Propellers |
| 7H | Alteration and Repair of Aircraft |
| 7J | Special Requirements for Air Line Aircraft |
| 14 | Relative Lift Distribution in Any Biplane |
| 14 | Requirements for Approved Type Certificates |
| 26 | Design Information for Aircraft |
| **Civil Air Regulations (CAR)** | |
| CAR 1 | Certification, Identification, and Marking of Aircraft and Related Products |
| CAR 2 | Aircraft Identification Mark |
| CAR 3 | Airplane Airworthiness—Normal, Utility, Acrobatic, and Restricted Purpose Categories |
| CAR 4a | Airplane Airworthiness |
| CAR 4b | Airplane Airworthiness: Transport Categories |
| CAR 6 | Rotorcraft Airworthiness: Normal Category |
| CAR 7 | Rotorcraft Airworthiness: Transport Categories |
| CAR 8 | Aircraft Airworthiness: Restricted Category |
| CAR 9 | Aircraft Airworthiness: Limited Category |
| CAR 10 | Certification and Approval of Import Aircraft and Related Products |
| CAR 13 | Aircraft Engine Airworthiness |
| CAR 14 | Aircraft Propeller Airworthiness |
| CAR 18 | Maintenance, Repair, and Alteration of Certificated Aircraft and of Aircraft Engines, Propellers and Instruments |
| CAR 40 | Scheduled Interstate Air Carrier Certification and Operation Rules |
| Special CAR 425-C | Provisional Certification and Operation of Aircraft |
| Special CAR 406 | Application of Transport Category Performance Requirements to C-46 Type Aircraft |
| **Civil Aeronautics Manual (CAM)** | |
| CAM 1 | Certification, Identification, and Marking of Aircraft and Related Products |
| CAM 2 | Production Certificates |
| CAM 3 | Airplane Airworthiness: Normal, Utility, and Acrobatic Categories |
| CAM 4a | Airplane Airworthiness |
| CAM 4b | Airplane Airworthiness: Transport Categories |
| CAM 6 | Rotorcraft Airworthiness |
| CAM 7 | Rotorcraft Airworthiness: Transport Categories |
| CAM 8 | Aircraft Airworthiness: Restricted Category |
| CAM 9 | Aircraft Airworthiness: Limited Category |
| CAM 10 | Certification and Approval of Import Aircraft and Related Products |
| CAM 13 | Aircraft Engine Airworthiness |
| CAM 14 | Aircraft Propeller Airworthiness |
| CAM 18 | Maintenance, Repair, and Alteration of Airframes, Powerplants, Propellers, and Appliances |

**Figure 2-11.** *Predecessor Regulations to the Federal Aviation Regulations (14 CFR).*

be missing. These numbers were either assigned to ACs still in preparation that will be issued at a later date or were assigned to ACs that have been canceled.

### *The AC Numbering System*

There are three parts to an AC number, as in 25-42-C.

- The first part of the number identifies the subject matter area of the AC. This corresponds to the part of the FAA's regulations. In the above example, this would be part 25.
- The second part of the number, beginning with the dash, is a sequential number within each subject area. In the above example, this would be the 42nd AC relating to part 25.
- The third part of the number is a letter assigned by the originating office showing the revision sequence if an AC is revised. The first version of an AC does not have a revision letter. In the above example, this is third revision, as designated by the "C." *[Figure 2-12]*

## Airworthiness Directives (AD)

In accordance with 14 CFR part 39, the FAA issues ADs in response to deficiencies and/or unsafe conditions found in aircraft, engines, propellers, or other aircraft parts. ADs require that the relevant problem must be corrected on all aircraft or aircraft parts using the same design. ADs are initiated as either proposed, corrective, or final (telegraphic) via the Federal Register. The Federal Register is the official daily publication of the United States Government. It is the printed method of informing the public of laws that are enacted or will be enacted. Electronic versions of ADs are available from the Federal Register and from the Regulatory and Guidance Library. You can search by manufacturer, model, or AD number. All ADs are "incorporated by reference" into part 39 and are considered final. ADs must be followed to remain in compliance with the FAA. Once an AD has been issued, a person/company is authorized to use the affected aircraft or part only if it has been corrected in accordance with the AD.

### *Types of Airworthiness Directives (AD)*

Three types of ADs are issued:

- Notice of Proposed Rulemaking (NPRM), followed by a Final Rule
- Final Rule; Request for Comments
- Emergency ADs

The standard AD process is to issue an NPRM followed by a Final Rule. After an unsafe condition is discovered, a proposed solution is published as an NPRM and solicits public comment on the proposed action. After the comment period closes, the final rule is prepared, taking into account all substantive comments received with the rule perhaps being changed as warranted by the comments. The preamble to the final rule AD provides response to the substantive comments or states there were no comments received.

In certain cases, the critical nature of an unsafe condition may warrant the immediate adoption of a rule without prior notice and solicitation of comments. This is an exception to the standard process. If time for the terminating action to be accomplished is too short to allow for public comment (that is, less than 60 days), then a finding of impracticability is justified for the terminating action, and it can be issued as an immediately adopted rule. The immediately adopted rule is published in the Federal Register with a request for comments. The Final Rule AD may be changed later if substantive comments are received.

An Emergency AD is issued when an unsafe condition exists that requires immediate action by an owner/operator. The intent of an Emergency AD is to rapidly correct an urgent safety deficiency. An Emergency AD may be distributed by fax, letter, or other methods. It is issued and effective to only the people who actually receive it. This is known as "actual notice." All known owners and operators of affected U.S.-registered aircraft, or those aircraft that are known to have an affected product installed, are sent a copy of an Emergency AD. To make the AD effective to all persons, a follow up publication of the Final Rule AD in the Federal Register is critical. This Final Rule AD must be identical to the Emergency AD and is normally published in the Federal Register within 30 days of the Emergency AD issue.

### *AD Content*

Generally, ADs include:

- A description of the unsafe condition
- The product that the AD applies to
- The required corrective action or operating limitations or both
- The AD effective date
- A compliance time
- Where to go for more information
- Information on alternative methods of compliance with the requirements of the AD

### *AD Number*

ADs have a three-part number designator. The first part is the calendar year of issuance. The second part is the biweekly period of the year when the number is assigned. The third part is issued sequentially within each biweekly period.

### *Applicability and Compliance*

The AD subject line specifically identifies the TC holder of the aircraft or products affected by the AD. The specific models affected and any special considerations, such as specific installed part numbers or modifications, are listed in the AD applicability section. In order to find all applicable ADs for a specific product, you must search for ADs on the product, aircraft, engine(s), propeller, or any installed appliance. If there are multiple series under the aircraft or engine model, you must also search for ADs applicable to the model, as well as the specific series of that model. The final determination of ADs applicable to a particular product can only be made by a thorough examination of the ADs and the product logbooks. No person may operate a product that an AD applies to, except in accordance with the requirements of the AD. Furthermore, the owner or operator of an aircraft is required by 14 CFR part 91, section 91.403 to maintain the aircraft in compliance with all ADs. The AD specifies a compliance time that relates to the effective date of the AD. That compliance time determines when the actions are required.

### *Alternative Method of Compliance*

Different approaches or techniques that are not specified in an AD can, after FAA approval, be used to correct an unsafe condition on an aircraft or aircraft product. Although the alternative was not known at the time the AD was issued, an alternative method may be acceptable to accomplish the intent of the AD. A compliance time that differs from the requirements of the AD can also be approved if the revised time period and approved alternative method provides an acceptable level of safety as the requirements of the AD.

## Special Airworthiness Information Bulletin (SAIB)

A Special Airworthiness Information Bulletin (SAIB) is an information tool that the FAA uses to alert, educate, and make recommendations to the aviation community. SAIBs contain non-regulatory information and guidance that does not meet the criteria for an AD. *[Figure 2-13]*

## Aircraft Specifications

Specifications were originated during implementation of the Air Commerce Act of 1926. Specifications are FAA recordkeeping documents issued for both type-certificated and non-type-certificated products that have been found eligible for U.S. airworthiness certification. Although they are no longer issued, specifications remain in effect and will be further amended. Specifications covering type-certificated products may be converted to a TCDS at the option of the TC holder. However, to do so requires the TC holder to provide an equipment list. A specification is not part of a TC. Specifications are subdivided into five major groups as follows:

1. Group I—Type Certificate Aircraft, Engines, and Propellers. Covering standard, restricted, and limited types issued for domestic, foreign, and military surplus products.
2. Group II—Aircraft, Engine, and Propeller Approvals. Covering domestic, foreign, and military surplus products constructed or modified between October 1, 1927, and August 22, 1938. All have met minimum airworthiness requirements without formal type certification. Such products are eligible for standard airworthiness certification as though they are type-certificated products.
3. Group III—Aircraft, Engine, and Propeller Approvals. Covering domestic products manufactured prior to October 1, 1927, foreign products manufactured prior to June 20, 1931, and certain military surplus engines and propellers. All have met minimum airworthiness requirements of the Air Commerce Act of 1926 and implementing Air Commerce Regulations without formal type certification. Such products are eligible for standard airworthiness certification as though they are type-certificated products.
4. Group IV—Engine Ratings. Covering unapproved engines rated for maximum power and speed only, their use being limited to specific aircraft with maximum gross weights less than 1,000 pounds. Such engines are not eligible for independent airworthiness certification. These ratings are no longer issued.
5. Group V—Engine Approvals. Covering military surplus engines meeting CAR 13 design requirements without formal type certification. Such engines are eligible for airworthiness certification as though they are type-certificated engines.

## Supplemental Type Certificates (STC)

When an aircraft is designed and that design is formally approved for manufacturing, the manufacturer is issued a Type Certificate (TC). The TC is issued by the FAA to signify the airworthiness of an aircraft design and may not be changed except by formal authorization of the FAA. This formal authorization supplements the original TC and is called the Supplemental Type Certificate (STC). Therefore, the STC issued by the FAA approves a product (aircraft, engine, or propeller) modification. *[Figure 2-14]* The STC defines the product design change, states how the modification affects the existing type design, and lists serial number effectivity. It also identifies the certification basis listing specific regulatory compliance for the design change. Information contained in the certification basis is helpful for those applicants proposing subsequent product modifications and evaluating certification basis compatibility with other STC modifications. Refer to *Figure 2-15* for a listing of how

## Advisory Circular Numbering System

**1. General.** The advisory circular numbers relate to the FAR subchapter title and correspond to the Parts, and when appropriate, to the specific sections of the Federal Aviation Regulations.

**2. General and specific subject numbers.** The subject numbers and related subject areas are as follows:

| General Subject Number (1) | Specific Subject Number (2) | Subject |
|---|---|---|
| **00** | | **GENERAL** |
| | 1 | Definitions and Abbreviations |
| **10** | | **PROCEDURAL RULES** |
| | 11 | General Rule-Making Procedures |
| | 13 | Investigation and Enforcement Procedures |
| **20** | | **AIRCRAFT** |
| | 21 | Certification Procedures for Products and Parts |
| | 23 | Airworthiness Standards: Normal, Utility, and Acrobatic Category Airplanes |
| | 25 | Airworthiness Standards: Transport Category Airplanes |
| | 27 | Airworthiness Standards: Normal Category Rotorcraft |
| | 29 | Airworthiness Standards: Transport Category Rotorcraft |
| | 31 | Airworthiness Standards: Manned Free Balloons |
| | 33 | Airworthiness Standards: Aircraft Engines |
| | 34 | Fuel Venting and Exhaust Emission Requirements for Turbine Engine Powered Airplanes |
| | 35 | Airworthiness Standards: Propellers |
| | 36 | Noise Standards: Aircraft Type and Airworthiness Certification |
| | 39 | Airworthiness Directives |
| | 43 | Maintenance, Preventive Maintenance, Rebuilding and Alteration |
| | 45 | Identification and Registration Marking |
| | 47 | Aircraft Registration |
| | 49 | Recording of Aircraft Titles and Security Documents |
| **60** | | **AIRMEN** |
| | 61 | Certification: Pilots and Flight Instructors |
| | 63 | Certification: Flight Crewmembers Other Than Pilots |
| | 65 | Certification: Airmen Other Than Flight Crewmembers |
| | 67 | Medical Standards and Certification |
| **70** | | **AIRSPACE** |
| | 71 | Designation of Federal Airways, Area Low Routes, Controlled Airspace, and Reporting Points |
| | 73 | Special Use Airspace |
| | 75 | Establishment of Jet Routes and Area High Routes |
| | 77 | Objects Affecting Navigable Airspace |
| **90** | | **AIR TRAFFIC AND GENERAL OPERATING RULES** |
| | 91 | General Operating and Flight Rules |
| | 93 | Special Air Traffic Rules and Airport Traffic Patterns |
| | 95 | IFR Altitudes |
| | 97 | Standard Instrument Approach Procedures |
| | 99 | Security Control of Air Traffic |
| | 101 | Moored Balloons, Kites, Unmanned Rockets and Unmanned Free Balloons |
| | 103 | Ultralight Vehicles |
| | 105 | Parachute Jumping |
| | 107 | Airport Security |
| | 108 | Airplane Operators Security |
| | 109 | Indirect Air Carrier Security |
| **119** | | **CERTIFICATION: AIR CARRIERS AND COMMERCIAL OPERATORS** |
| **120** | | **AIR CARRIERS, AIR TRAVEL CLUBS, AND OPERATORS FOR COMPENSATION OR HIRE: CERTIFICATION AND OPERATIONS** |
| | 121 | Certification and Operations: Domestic, Flag, and Supplemental Air Carriers and Commercial Operators of Large Aircraft |
| | 125 | Certification and Operations: Airplanes Having a Seating Capacity of 20 or More Passengers or a Maximum Payload Capacity of 6,000 Pounds or More |
| | 127 | Certification and Operations of Scheduled Air Carriers with Helicopters |
| | 129 | Operations of Foreign Air Carriers |
| | 133 | Rotorcraft External-Load Operations |
| | 135 | Air Taxi Operators and Commercial Operators |
| | 137 | Agricultural Aircraft Operations |
| | 139 | Certification and Operations: Land Airports Serving CAB-Certificated Air Carriers |
| **140** | | **SCHOOLS AND OTHER CERTIFICATED AGENCIES** |
| | 141 | Pilot Schools |
| | 143 | Ground Instructors |
| | 145 | Repair Stations |
| | 147 | Aviation Maintenance Technician Schools |
| **150** | | **AIRPORT NOISE COMPATIBILITY PLANNING** |
| | 151 | Federal Aid to Airports |
| | 152 | Airport Aid Program |

**Figure 2-12.** *List of advisory circular numbers.*

## Advisory Circular Numbering System

| General Subject number (1) | Specific Subject Number (2) | Subject |
|---|---|---|
| | 155 | Release of Airport Property from Surplus Property Disposal Restrictions |
| | 156 | State Block Grant Pilot Program |
| | 157 | Notice of Construction, Alteration, Activation, and Deactivation of Airports |
| | 158 | Passenger Facilities Charges |
| | 159 | National Capital Airports |
| | 159/10 | Washington National Airport |
| | 159/20 | Dulles International Airport |
| | 161 | Notice and Approval of Airport Noise and Access Restrictions |
| | 169 | Expenditures of Federal Funds for Nonmilitary Airports or Air Navigational Facilities Thereon |
| **170** | | **NAVIGATIONAL FACILITIES** |
| | 170 | Establishment and Discontinuance Criteria for Airport Traffic Control Tower Facilities |
| | 171 | Non-Federal Navigation Facilities |
| **180** | | **ADMINISTRATIVE REGULATIONS** |
| | 183 | Representatives of the Administrator |
| | 185 | Testimony by Employees and Production of Records in Legal Proceedings |
| | 187 | Fees |
| | 189 | Use of Federal Aviation Administration Communication System |
| **190** | | **WITHHOLDING SECURITY INFORMATION** |
| | 191 | Withholding Security Information from Disclosure Under the Air Transportation Security Act of 1974 |
| | 198 | Aviation Insurance Program |
| **210** | | **FLIGHT INFORMATION** |
| | 211 | Aeronautical Charts and Flight Information Publications |
| | 212 | Publication Specification: Charts and Publications |
| **400** | | **COMMERCIAL SPACE TRANSPORTATION** |
| | 440 | Financial Responsibility |

1—Based on Federal Aviation Regulation Subchapter Titles (Excluding the 210 series).

2—Based on Federal Aviation Regulation Part Titles (Excluding the 210 series).

**3. Within the General Subject Number Areas, Specific selectivity in advisory circular mail lists is available corresponding to the applicable FAR Parts. For example: under the 60 general subject area, separate mail lists for advisory circulars issued in the 61, 63, 65, or 67 series are available. An AC numbered "60" goes to all numbers in the 60 series. When the volume of circulars in a series warrants a sub-subject breakdown, the general number is followed by a slash and a sub-subject number. Material in the 150 series, Airports, is issued under the following sub-subjects:**

| | |
|---|---|
| **150/5000** | Airport Planning. |
| **150/5020** | Noise Control and Compatibility Planning for Airports. |
| **150/5100** | Federal-aid Airport Program. |
| **150/5150** | Surplus Airport Property Conveyance Programs. |
| **150/5190** | Airport Compliance Program. |
| **150/5200** | Airport Safety–General. |
| **150/5210** | Airport Safety Operations (Recommended Training, Standards, Manning). |
| **150/5220** | Airport Safety Equipment and Facilities. |
| **150/5230** | Airport Ground Safety System. |
| **150/5240** | Civil Airports Emergency Preparedness. |
| **150/5325** | Influence of Aircraft Performance on Aircraft Design. |
| **150/5335** | Runway, Taxiway, and Apron Characteristics. |
| **150/5340** | Airport Visual Aids. |
| **150/5345** | Airport Lighting Equipment. |
| **150/5360** | Airport Buildings. |
| **150/5370** | Airport Construction. |
| **150/5380** | Airport Maintenance. |
| **150/5390** | Heliports. |

**4. Individual circular identification numbers. Each circular has a subject number followed either by a dash and a consecutive number (135-15) or a period with a specific FAR section number, followed by a dash and a consecutive number (135.169-2) identifying the individual circular. This consecutive number is not used again in the same subject series. Revised circulars have a letter A, B, C, etc., after the consecutive number to show complete revisions. Changes to circulars have Chg. 1, Chg. 2, Chg. 3, etc., after the identification number on pages that have been changed. The Date on a revised page is changed to the date of the Change transmittal.**

**Figure 2-12.** *List of advisory circular numbers (continued).*

TCs and STCs are numbered.

Possession of the STC document does not constitute rights to the design data or installation of the modification. The STC and its supporting data (drawings, instructions, specifications, and so forth) are the property of the STC holder. You must contact the STC holder to obtain rights for the use of the STC.

### Type Certificate Data Sheets (TCDS)

The TCDS is a formal description of the aircraft, engine, or propeller. It lists limitations and information required for type certification including airspeed limits, weight limits, thrust limitations, and so forth.

TCDSs and specifications set forth essential factors and other conditions that are necessary for U.S. airworthiness certification. Aircraft, engines, and propellers that conform to a U.S. TC are eligible for U.S. airworthiness certification when found to be in a condition for safe operation and ownership requisites are fulfilled. *[Figure 2-16]*

TCDSs were originated and first published in January 1958. Title 14 of the CFR part 21, section 21.41 indicates they are part of the TC. As such, a TCDS is evidence the product has been type certificated. Generally, TCDSs are compiled from details supplied by the TC holder; however, the FAA may request and incorporate additional details when conditions warrant. *[Figure 2-17]*

Under federal law, no civil aircraft registered in the United States can operate without a valid airworthiness certificate. This certificate must be approved and issued by the FAA; and it is only issued if the aircraft and its engines, propellers, and appliances are found to be airworthy and meet the requirements of an FAA-approved TC. The FAA issues a TC when a new aircraft, engine, propeller, and so forth, is found to meet safety standards set forth by the FAA. The TCDS lists the specifications, conditions, and limitations that the airworthiness requirements were met under for the specified product, such as engine make and model, fuel type, engine limits, airspeed limits, maximum weight, minimum crew, and so forth. TCDSs are issued and revised as necessary to accommodate new models or other major changes in the certified product. TCDSs are categorized by TC holder and product type.

## FAA Handbooks & Manuals

The FAA publishes handbooks and manuals for beginners and aviation professionals. Publications are updated periodically to reflect new FAA regulations and technical developments. *Figure 2-18* shows a list of aircraft and aviation handbooks and manuals available on the FAA website (www.faa.gov).

## Non-FAA Documents

### Air Transport Association ATA iSpec 2200

To standardize the technical data and maintenance activities on large and therefore complex aircraft, the ATA e-Business Program has established a classification of maintenance related actions. These are arranged with sequential numbers assigned to ATA chapters. These chapters are consistent regardless of the large aircraft that is being worked on. *[Figure 2-19]*

### Manufacturers' Published Data

The original equipment manufacturer (OEM) is usually the best source of information for the operation of and maintenance on a particular product. If the product is a TC'd or STC'd item, 14 CFR part 21, section 21.50 requires the holder of the design approval to provide one set of complete ICAs. Additional requirements for ICAs are specified in sections 23.1529, 25.1529, 27.1529 and 29.1529. These sections further refer the reader to 14 CFR part 23, Appendix A; part 25, Appendix H; part 27, Appendix A; and part 29, Appendix A. Regardless of the appendix referred to, the requirements in the appendix for the ICA are as follows:

- General: The aircraft ICA must contain instructions for continued airworthiness for each engine, propeller, or appliance and the interface of those appliances and products with the aircraft.
- Format: The ICA must be in the form of a manual or manuals appropriate to the data being provided.
- Content: The manual contents must be in English and must include the following:
    - Introductory information, including an explanation of the airplane's features and data as necessary to perform maintenance or preventive maintenance
    - A description of the aircraft and its systems, including engine, propeller, and appliances
    - Basic operating information describing how the aircraft and its components are controlled
    - Servicing information with such detail as servicing parts, tank capacities, types of fluid to be used, applicable pressures for the various systems, access panels for inspection and servicing, lubrication points, and types of lubricants to be used

The maintenance instructions must include the following data:

- Recommended schedule for cleaning, inspecting, adjusting, testing, and lubricating the various parts
- Applicable wear tolerances

**FAA**
**Aviation Safety**

# SPECIAL AIRWORTHINESS INFORMATION BULLETIN

**SAIB:** CE-15-13
**Date:** April 15, 2015

**SUBJ:** FUSELAGE – Seat Belt Mounting Bracket

*This is information only. Recommendations aren't mandatory.*

## Introduction

This Special Airworthiness Information Bulletin is to alert owners, operators, and maintenance technicians of an airworthiness concern with aluminum seat belt mounting brackets affecting all **Cessna Models 120 and 140** airplanes. Textron Aviation has issued Service Bulletin SEB-25-03, dated February 17, 2015, to address this concern.

At this time, the airworthiness concern is not an unsafe condition that would warrant airworthiness directive (AD) action under Title 14 of the Code of Federal Regulations (14 CFR) part 39.

## Background

On July 5, 2014, an accident occurred in Parma, New York where the pilot seat belt mounting bracket, part number (p/n) 0425132, failed after the airplane overturned following departure from the runway. Although cause of the failed bracket has not been determined and the investigation is ongoing, it was noted that the original Cessna seat belt installation had been replaced with a four-point Aero Fabricators harness per Supplemental Type Certificate (STC) SA1429GL in 2003. The failed bracket was made of aluminum. However, Cessna now only provides steel brackets as a replacement part for the aluminum brackets.

## Recommendations

The FAA recommends that owners, operators, and maintenance personnel of the affected airplanes replace aluminum brackets with steel brackets following Cessna Service Bulletin SEB-25-03 dated February 17, 2015. To make the determination as to whether the bracket is made of aluminum or steel, a magnet may be used or look for evidence of iron oxide (rust).

## For Further Information Contact

Gary D. Park, Aerospace Engineer, ACE-118W phone: (316) 946-4123; fax: (316) 946-4107; e-mail: gary.park@faa.gov.

## For Related Service Information Contact

Cessna Aircraft Company, Customer Support Service, P.O. Box 7706, Wichita, Kansas; telephone: (316) 517-5800; fax: (316) 517-7271.

SAMPLE

**Figure 2-13.** *Special Airworthiness Information Bulletin (SAIB).*

United States Of America

Department of Transportation - Federal Aviation Administration

# Supplemental Type Certificate

*Number* SA7855SW

*This Certificate issued to* Commander Premier Aircraft Corporation
20 Stanford Drive
Farmington, CT 06032

*certifies that the change in the type design for the following product with the limitations and conditions therefor as specified hereon meets the airworthiness requirements of Part* 23 *of the* Federal Aviation Regulations.

*Original Product Type Certificate Number:* A12SO

*Make:* Commander

*Model:* 114

*Description of Type Design Change:*
Installation of McCauley B3D32C419/82NHA-5 Propeller on Commander Model 114 airplane in accordance with Commander Aircraft Co., Installation Instructions dated July 13, 1990, Revision A dated August 22, 1990, or later FAA approved revision.

*Limitations and Conditions:*
FAA approved Commander Aircraft Co. Flight Manual Supplement dated August 23, 1990, must accompany this modification. The installer must determine whether this design change is compatible with previously approved modifications. If the holder agrees to permit another person to use this certificate to alter a product, the holder must give the other person written evidence of that permission.

*This certificate and the supporting data which is the basis for approval shall remain in effect until surrendered, suspended, revoked or a termination date is otherwise established by the Administrator of the Federal Aviation Administration.*

*Date of application:* February 09, 1990

*Date of issuance:* August 23, 1990

*Date reissued:* March 03, 2006

*Date amended:* Amd. 1, September 10, 1990

*By direction of the Administrator*

(*Signature*)
Michele M. Owsley, Manager
Airplane Certification Office,
Southwest Region

(*Title*)

*Any alteration of this certificate is punishable by a fine of not exceeding $1,000, or imprisonment not exceeding 3 years, or both.*

FAA Form 8110-2(10-68) Page 1 of 1 *This certificate may be transferred in accordance with FAR 21.47.*

SAMPLE

**Figure 2-14.** *Supplemental type certificate (STC).*

- Recommended overhaul periods
- Details for an inspection program that identifies both the frequency and the extent of the inspections necessary to provide for continued airworthiness
- Troubleshooting information
- The order and method for proper removal and replacement of parts
- Procedures for system testing during ground operations
- Diagrams for structural access plates
- Details for application of special inspection techniques
- Information concerning the application of protective treatments after inspection
- Information relative to the structural fasteners
- List of any special tools needed

### Airworthiness Limitations

The ICA must contain a separate and clearly distinguishable section titled "Airworthiness Limitations." Within this section are mandatory replacement times, structural inspection interval, and related inspection procedures.

All of this is included in the initial release of documents when the aircraft is delivered. However, over the course of the life of an aircraft, various modifications can and often do occur. Whether these are as simple as a new cabin to galley sliding door, or as complex as a navigation related STC, any major alteration requires that this type of maintenance data be provided to the owner, so that subsequent maintenance, inspection, and repair can be properly accomplished. As aircraft and their systems become more and more complex, and society continues its preoccupation with litigation for every incident, it is imperative that the technician have the right information, that it is current, and that they have the proper tools, including those required for any special inspection, and correct replacement parts. If any one of these items is required, and the technician does not have it accessible, they are in violation of 14 CFR sections 65.81(b), 43.13(a), and 43.16 if they attempt to return the aircraft to service.

Manufacturers may provide this required information in a variety of different manuals:

- Operating Instructions—The Airplane Flight Manual (AFM) or the Pilot's Operating Handbook (POH) provides the pilot with the necessary information to properly operate the aircraft. These manuals are usually listed in the aircraft TCDS, and therefore are a required item for the aircraft to be considered airworthy. Note that the AFM is generally serial number specific, whereas the POH is model specific. After 1978, the POHs generally took on both roles.
- Maintenance Manuals—These manuals are often referred to as Aircraft Maintenance Manual (AMM) or Component Maintenance Manual (CMM).

The AMM is focused on the entire aircraft and provides the "big picture" for the maintenance technician. It provides information concerning the maintenance, including troubleshooting and repair, of the aircraft and systems on the aircraft.

The CMM, on the other hand, is focused on a specific item or component, such as hydraulic pump, generator, or thrust reverser. It provides the bench mechanic with detail troubleshooting information and usually serves as an overhaul manual giving details for disassembly, cleaning, inspection, repair as necessary, reassembly, and testing in accordance with approved standards and technical data accepted by the Administrator. Refer to 14 CFR part 43 section 43.2(a). When maintenance is done according to the CMM, the technician must always include the appropriate references in the maintenance record entry required by 14 CFR part 43, section 43.9 or 43.11.

### Service Bulletins (SB)

Throughout the life of a product (whether TC'd or not), manufacturing defects, changes in service, or design improvements often occur. When that happens, the OEM frequently uses an SB to distribute the information to the operator of the aircraft. SBs are good information and should be strongly considered by the owner for implementation to the aircraft. However, SBs are not required unless they are referred to in an AD note or if compliance is required as a part of the authorized inspection program. Refer to section 14 CFR part 39, 39.27.

### Structural Repair Manual (SRM)

As the name implies, this manual carries detail information for the technician concerning an aircraft's primary and secondary structure, criteria for evaluating the severity of the detected damage, determining the feasibility of a repair, and alignment/inspection information. This manual is usually a separate manual for large aircraft. On small aircraft, this information is often included in the AMM.

## Forms

### Airworthiness Certificates

In addition to the registration certificate that indicates the ownership of an aircraft, an airworthiness certificate indicates the airworthiness of the aircraft. AC 21-12, Application for U.S. Airworthiness Certificate, FAA Form 8130-6, is a comprehensive guide for the completion of the application form for this certificate. There are two certificates: standard

## FAA Project Numbering and Designators

**FAA Project Numbers will use the following format:**

AAnnnnnYY-X

Where:

- AA is the two-letter designator for Project Type – see table 1 below
- nnnnn is the integer sequential number for the specified ACO; e.g., 00146
- YY is the two-letter designator for the Aircraft Certification Office (ACO) – see Table 2 below
- X is the one-letter designator for the Product Type – see Table 3 below

As an example, TC00125AT-A would be a TC project assigned by the Atlanta ACO on a small airplane with the assigned number 00125.

**Table 1–Project Type Designators**

| Code | Description |
|---|---|
| TC | New Type Certificate (TC) |
| ST | New Supplemental Type Certificate (STC) |
| AT | Amended Type Certificate |
| SA | Amended Supplemental Type Certificate |
| SP | Special Project (e.g. approval under section 21.305 project) |
| PM | Parts Manufacturer Approval (PMA) |

**Table 2–Aircraft Certification Office (ACO) Designators**

| Code | Branch | Description |
|---|---|---|
| AC | ASW-150 | Ft. Worth Airplane Certification Office |
| AK | ACE-115N | Anchorage Aircraft Certification Office |
| AT | ACE-115A | Atlanta Aircraft Certification Office |
| BA | ANM-100B | Boeing Aviation Safety Oversight Office (BASOO) |
| BO | ANE-150 | Boston Aircraft Certification Office |
| CE | ACE-112 | Small Airplane Directorate |
| CH | ACE-115C | Chicago Aircraft Certification Office |
| DE | ANM-100D | Denver Aircraft Certification Office |
| EN | ANE-140 | Engine Certification Office |
| GU | ACE-100G | Gulfstream Aviation Safety Oversight Office (GASOO) |
| IB | ANM-116 | Transport Airplane Directorate International Branch |
| LA | ANM-100L | Los Angeles Aircraft Certification Office |
| MC | ACE-100M | Military Certification Office |
| NY | ANE-170 | New York Aircraft Certification Office |
| RC | ASW-170 | Ft. Worth Rotorcraft Certification Office |
| SC | ASW-190 | Ft. Worth Special Certification Office |
| SE | ANM-100S | Seattle Aircraft Certification Office |
| WI | ACE-115W | Wichita Aircraft Certification Office |

**Table 3–Product Type Designators**

| Code | Description |
|---|---|
| A | Small Airplane |
| B | Balloon |
| E | Engine |
| G | Glider |
| P | Propeller |
| R | Rotorcraft |
| S | Airship |
| T | Transport Airplane |
| I | Experimental |
| Q | Other, or not product |

**Table 4–Directorate Designators**

| Code | Description |
|---|---|
| EPD | Engine-Propeller |
| RCD | Rotorcraft |
| SAD | Small Airplane |
| TAD | Transport Airplane |

**Table 5–Aircraft Evaluation Group Designators**

| Code | Description |
|---|---|
| BOS-AEG | Boston AEG |
| FTW-AEG | Fort Worth AEG |
| MKC-AEG | Kansas City AEG |
| LGB-AEG | Long Beach AEG |
| SEA-AEG | Seattle AEG |

As an example, TC00125AT-A would be a TC project assigned by the Atlanta ACO on a small airplane with the assigned number 00125.

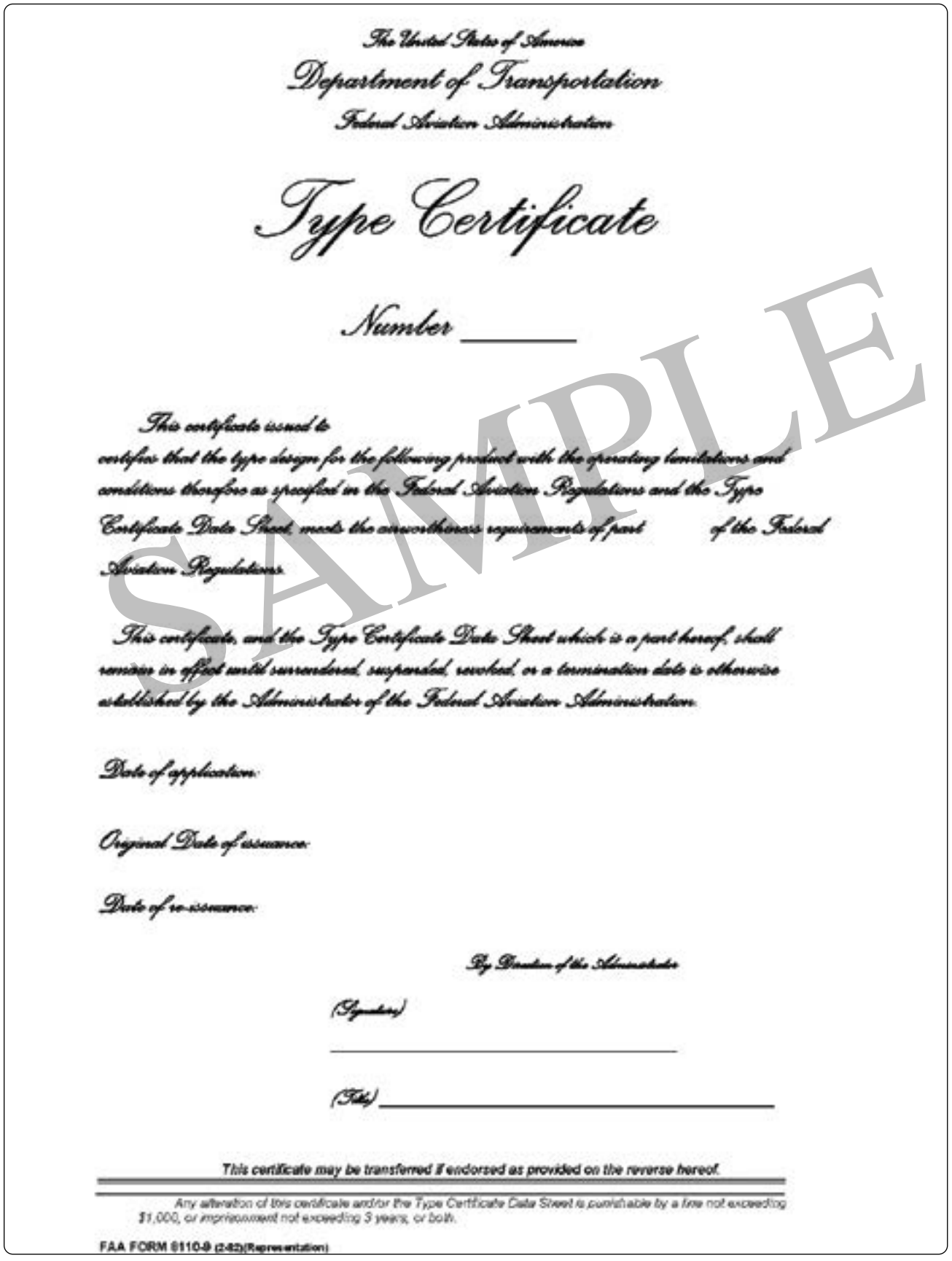
**Figure 2-15.** *Numbering system for type certificates (TCs) and supplemental type certificates (STCs).*

and special. FAA Form 8100-2, Standard Airworthiness Certificate, may be issued to allow operation of a type-certificated aircraft in one or more of the following categories: *[Figure 2-20]*

- Normal
- Utility
- Acrobatic
- Commuter
- Transport
- Manned free balloon
- Special classes

FAA Form 8130-7, Special Airworthiness Certificate, may be issued to authorize the operation of an aircraft in the following categories: *[Figure 2-21]*

- Primary
- Restricted
- Multiple
- Limited
- Light-sport



The United States of America

Department of Transportation

Federal Aviation Administration

# Type Certificate

Number _____

This certificate issued to

certifies that the type design for the following product with the prescribed limitations and conditions thereto as specified in the Federal Aviation Regulations and the Type Certificate Data Sheet, meets the airworthiness requirements of part of the Federal Aviation Regulations.

This certificate, and the Type Certificate Data Sheet which is a part hereof, shall remain in effect until surrendered, suspended, revoked, or a termination date is otherwise established by the Administrator of the Federal Aviation Administration.

Date of application:

Original Date of issuance:

Date of re-issuance:

By Direction of the Administrator

(Signature)

_____

(Title) _____

This certificate may be transferred if endorsed as provided on the reverse hereof.

Any alteration of this certificate and/or the Type Certificate Data Sheet is punishable by a fine not exceeding $1,000, or imprisonment not exceeding 3 years, or both.

FAA FORM 8110-9 (2-82)(Representation)

SAMPLE

**Figure 2-16.** *Type certificate.*

**DEPARTMENT OF TRANSPORTATION**
**FEDERAL AVIATION ADMINISTRATION**

A24CE
Revision 111
Beechcraft

| | |
|---|---|
| 200 | A100-1 (U-21J) |
| 200C | A200 (C-12A) |
| 200CT | A200 (C-12C) |
| 200T | A200C (UC-12B) |
| B200 | A200CT (C-12D) |
| B200C | A200CT (FWC-12D) |
| B200CT | A200CT (C-12F) |
| B200T | A200CT (RC-12D) |
| 300 | A200CT (RC-12G) |
| 300LW | A200CT (RC-12H) |
| B300 | A200CT (RC-12K) |
| B300C | A200CT (RC-12P) |
| B300C (MC-12W) | A200CT (RC-12Q) |
| B300C (UC-12W) | B200C (C-12F) |
| 1900 | B200C (UC-12M) |
| 1900C | B200C (C-12R) |
| 1900C (C-12J) | B200C (UC-12F) |
| 1900D | B200GT |
| | B200CGT |

July 21, 2015

**TYPE CERTIFICATE DATA SHEET NO. A24CE**

This data sheet which is part of Type Certificate No. A24CE prescribes conditions and limitations under which the product for which the type certificate was issued meets the airworthiness requirements of the Federal Aviation Regulations.

Type Certificate Holder: Beechcraft Corporation
10511 E. Central
Wichita, Kansas 67206

Type Certificate Holder Record: Beech Aircraft Corporation transferred to
Raytheon Aircraft Company on April 15, 1996

Raytheon Aircraft Company transferred to
Hawker Beechcraft Corporation on March 26, 2007

Hawker Beechcraft Corporation transferred to
Beechcraft Corporation on April 12, 2013

**I. Model 200, Super King Air (Normal Category), Approved December 14, 1973 (See NOTES 10 and 11)**
**Model A200C (UC-12B), Super King Air (Normal Category), Approved February 21, 1979 (See NOTE 11)**
**Model 200C, Super King Air (Normal Category), Approved February 21, 1979 (See NOTE 11)**
**Model B200, Super King Air (Normal Category), Approved February 13, 1981 (See NOTES 10 and 11)**
**Model B200C, Super King Air (Normal Category), Approved February 13, 1981 (See NOTES 10 and 11)**
**Model B200C (C-12F), (UC-12F), (UC-12M) and (C-12R), Super King Air (Normal Category), Approved February 13, 1981, (See NOTES 10, 11, and 12)**
For Notes, refer to Data Pertinent to All Model 200 Series

Engine: Two United Aircraft of Canada, Ltd., or Pratt & Whitney PT6A-41 (turboprop) per Beech Specification BS 22096 (200, 200C, A200C)

| Page No | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *Rev. No.* | *111* | *101* | *104* | *97* | *82* | *97* | *82* | *99* | *97* | *101* | *111* | *104* | *96* | *101* | *101* | *110* |
| Page No | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 |
| *Rev. No.* | *111* | *101* | *82* | *95* | *91* | *108* | *111* | *101* | *101* | *101* | *101* | *101* | *100* | *78* | *101* | *110* |
| Page No | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | | | | | |
| *Rev. No.* | *104* | *111* | *110* | *110* | *111* | *108* | *111* | *96* | *100* | *101* | *101* | | | | | |

**Figure 2-17.** *Type Certificate Data Sheet.*

| Aircraft |
|---|
| Aircraft Weight and Balance Handbook (FAA-H-8083-1A) |
| Airplane Flying Handbook (FAA-H-8083-3A) |
| IR-M 8040-1C, Airworthiness Directives Manual |
| Amateur-built Aircraft & Ultralight Flight Testing Handbook |
| Aviation Maintenance Technician Handbook–General (FAA-H-8083-30) |
| Aviation Maintenance Technician Handbook–Airframe (FAA-H-8083-31) |
| Aviation Maintenance Technician Handbook–Powerplant (FAA-H-8083-32) |
| Balloon Flying Handbook (FAA-H-8083-11A) |
| Glider Flying Handbook (FAA-H-8083-13A) |
| Parachute Rigger Handbook (FAA-H-8083-17) |
| Rotorcraft Flying Handbook (FAA-H-8083-21) |

| Aviation |
|---|
| Advanced Avionics Handbook (FAA-H-8083-6) |
| Aerodynamics for Navy Aviators (NAVAIR 00-80T-80) |
| Aeronautical Information Manual |
| Air Quality Handbook |
| Airship Pilot Manual |
| Airship Aerodynamics Technical Manual |
| Aviation Instructor's Handbook (FAA-H-8083-9A) |
| Balloon Safety Tips: False Lift, Shear, and Rotors (FAA-P-8740-39) |

| Aviation |
|---|
| Balloon Safety Tips: Powerlines & Thunderstorms (FAA-P-8740-34) |
| Banner Tow Operations (FAA/fs-i-8700-1) |
| Flight Navigator Handbook (FAA-H-8083-18) |
| Helicopter Flying Handbook (FAA-H-8083-21A) |
| Helicopter Instructor's Handbook (FAA-H-8083-4) |
| Instrument Flying Handbook (FAA-H-8083-15B) |
| Instrument Procedures Handbook (FAA-H-8083-16) |
| International Flight Information Manager |
| MC-4 Ram Air Free-fall Personnel Parachute System Technical Manual |
| Pilot Safety Brochures |
| Pilot's Handbook of Aeronautical Knowledge (FAA-H-8083-25A) |
| Plane Sense–General Aviation Information (FAA-H-8083-19A) |
| Risk Management Brochures |
| Risk Management Handbook (FAA-H-8083-2) |
| Safety Risk Management |
| Seaplane, Skiplane, and Float/Ski Equipped Helicopter Operations Handbook (FAA-H-8083-23) |
| Student Pilot Guide (FAA-H-8083-27A) |
| Tips on Mountain Flying (FAA-P-8740-60) |
| Weight-Shift Control Aircraft Flying Handbook (FAA-H-8083-5) |

| Examiner and Inspector |
|---|
| Flight Standards Information Management System (FSIMS) (FAA Order 8900.1) |
| Designee Management Handbook |
| Guide for Aviation Medical Examiners |
| General Aviation Airman Designee Handbook |

**Figure 2-18.** *FAA handbooks and manuals.*

- Experimental
- Special flight permit
- Provisional

Airworthiness certificates may be issued by either FAA personnel or FAA designees. Refer to 14 CFR part 183, sections 183.31 and 183.33. The certificate must not only be on board the aircraft (14 CFR part 91, section 91.203(a)(1)), but must also be "displayed at the cabin or flight deck entrance so that it is legible to the passengers or crew" 14 CFR part 91, (section 91.203(b)). Since the ability to obtain this certificate is based upon the requirement to inspect the aircraft to determine that it conforms to type design and is in condition for safe operation, it can also be revoked by the FAA if either of those two requirements ceases to exist.

### Aircraft Registration

Aircraft must be registered in the United States if the aircraft is not registered under the laws of a foreign country and is owned by either a citizen of the United States, a foreign citizen lawfully admitted to the United States, or a corporation organized in and doing business under U.S. laws and primarily based in the United States. This registration is accomplished by using FAA Form 8050-1, Aircraft Registration Application. The aircraft registration form is available online at www.faa.gov. The aircraft owner can mail in completed copy, and keep a copy of the form as temporary authority to operate the aircraft after the fee and evidence of ownership have been mailed or delivered to the Registry. When carried in the aircraft with an appropriate current airworthiness certificate or a special flight permit, a copy of this completed application provides authority to operate the aircraft in the United States for up to 90 days.

| Joint Aircraft Systems Component (JASC)/ATA Code Table | | |
|---|---|---|
| **Aircraft** | 35 Oxygen | **Powerplant System** |
| 11 Placards and Markings | 36 Pneumatic | 71 PowerPlant |
| 12 Servicing | 37 Vacuum | 72 Turbine/Turboprop Engine |
| 14 Hardware | 38 Water/Waste | 73 Engine Fuel and Control |
| 18 Helicopter Vibration | 45 Central Maintenance System (CMS) | 74 Ignition |
| **Airframe Systems** | 49 Airborne Auxiliary Power | 75 Air |
| 21 Air Conditioning | 51 Standard Practices/Structures | 76 Engine Control |
| 22 Auto Flight | 52 Doors | 77 Engine Indicating |
| 23 Communications | 53 Fuselage | 78 Engine Exhaust |
| 24 Electrical Power | 54 Nacelles/Pylons | 79 Engine Oil |
| 25 Equipment/Furnishings | 55 Stabilizers | 80 Starting |
| 26 Fire Protection | 56 Windows | 81 Turbocharging |
| 27 Flight Controls | 57 Wings | 82 Water Injection |
| 28 Fuel | **Propeller/Rotor Systems** | 83 Accessory Gearboxes |
| 29 Hydraulic Power | 61 Propellers/Propulsors | 85 Reciprocating Engine |
| 30 Ice and Rain Protection | 62 Main Rotor | |
| 31 Instruments | 63 Main Rotor Drive | |
| 32 Landing Gear | 64 Tail Rotor | |
| 33 Lights | 65 Tail Rotor Drive | |
| 34 Navigation | 67 Rotors Flight Control | |

**Figure 2-19.** *Maintenance classification.*

In addition to the completed application form, the owner must also submit evidence of ownership (such as a bill of sale) and a registration fee. A successful review of the application results in the issuance of AC Form 8050-3, Certificate of Aircraft Registration. (Note the AC prefix.)

14 CFR section 91.203(a)(2) requires that either the pink copy of the application or the actual certificate of registration be on board the aircraft during its operation.

If the registration is ever lost or damaged, it may be replaced by contacting the FAA Aircraft Registration Branch and providing them with the aircraft specific data, including make, model, N-number, and serial number. A replacement certificate fee and an explanation of the reason for the replacement certificate are also required.

### Radio Station License

A radio station license is required if the aircraft is equipped with radios, and the aircraft is planned to be flown outside the boundaries of the United States. A radio station license is not required for aircraft that are operated domestically. (A major change occurred on February 8, 1996, when the telecommunications Act of 1996 was signed into law.)

The Federal Communications Commission (FCC) formerly required that any communication transmitter installed in aircraft be licensed. These FCC licenses were valid for 5 years. This is not an FAA requirement. FAA inspectors who conducted ramp inspections and detected an expired radio station license were not required to notify the FCC, nor could they issue a violation to the owner/operator. Simply informing the operator of the expired radio station license was their only responsibility.

FSGA 96-06, a Flight Standards Information Bulletin (FSIB) for General Aviation (FSGA) titled "Elimination of Aircraft Radio Station Licenses" became effective on July 8, 1996. Although that FSIB had an effectivity of only 1 year, the elimination of the requirement for aircraft used only in domestic operations continues.

### FAA Form 337—Major Repair and Alteration

Refer to the current issue of AC 43.9-1, Instructions for Completion of FAA Form 337 for help completing FAA Form 337, Major Repair and Alteration (Airframe, Powerplant, Propeller, or Appliance). *[Figure 2-22]*

As the name clearly states, this form is to be used whenever major repairs or alterations are accomplished on an aircraft. The only exception would be that 14 CFR part 43, Appendix B, allows for a certificated repair station to RTS an aircraft after a major repair by using a signed and dated work order and a signed maintenance release.

- Information in item 1 comes directly from the aircraft dataplate, except for the tail number. That is to be compared to the aircraft registration form.

- Information in item 2 reflects the name and address listed on AC Form 8050-3, Certificate of Registration.
- Item 3 is used when there is no existing approved data for the intended repair or alteration. In that case, the technician can request that the local FSDO Principal Maintenance Inspector (PMI) review the data and then grant a field approval, shown by completing and signing this area. In many cases, this block is blank because the technician has found, used, and made reference to data already approved by the FAA.
- Item 4—If the repair or alteration is being done to the aircraft airframe, no entry is required since the data is identical to that in item 1. However, if the repair or alteration is being done to an engine, a propeller, or other appliance, entries must include the appropriate make, model, and serial number information.
- Item 5 should have "X" marked in either the "Repair" or the "Alteration" column.
- Item 6—Enter appropriate data as specified and check the proper box in B. The technician is encouraged to carefully read the preprinted statement in subparagraph D prior to signing this section.
- Item 7 must be completed by the IA or authorized individual from the repair station.
- Item 8 (on the reverse side) is for the description of the work accomplished. It must include a reference to the approved data used to conduct the required maintenance.

The form must be completed at least in duplicate, with the original provided to the owner/operator and a copy to the local FSDO within 48 hours of completing the maintenance and RTS. If the FAA Form 337 is used to document additional fuel tanks in the cabin or cargo, then an additional copy must be signed and in the aircraft at all times. Maintenance facilities and mechanics are encouraged to make a copy for their own records.

## Records

### Making Maintenance Record Entries

Title 14 of the CFR part 43, sections 43.9 and 43.11 require the technician to make appropriate entries of maintenance actions or inspection results in the aircraft maintenance record. How long those records must be kept is defined in 14 CFR part 91, section 91.417.

Whenever maintenance, preventive maintenance, rebuilding, or alteration work occurs on an aircraft, airframe, aircraft engine, propeller, appliance, or component part, a maintenance record entry must be created. The importance of compliance with this requirement cannot be overemphasized. Complete and organized maintenance logs for an aircraft can have significant (and usually positive) effect during the buy/sell negotiations of an aircraft. On the other hand, poorly organized and incomplete logs can have a detrimental effect upon the selling price of an aircraft.

### Temporary Records—14 CFR Part 91 Section 91.417(a)(1) and (b)(1)

These are records that must be kept by the owner until the work is repeated, superseded, or 1 year has transpired since the work was performed. These are typically records referring to maintenance, preventive maintenance, alteration, and all inspections. They include a description of the work performed (or reference to the FAA-accepted data); the date of completion; and the name, signature and certificate number of the person doing the RTS.

### Permanent Records—14 CFR Part 91, Section 91.417(a)(2) and (b)(2)

These records must be retained by the owner during the time they operate the aircraft. They are transferred with the aircraft at the time of sale. Typically, these are documents relating to total time in service, current status of life-limited parts, time since last overhaul, current inspection status, current status of applicable AD notes, and major alteration forms as required by 14 CFR part 43, section 43.9.

### Electronic Records

During the last 25 years, the field of aviation maintenance has seen a significant change in the documentation requirements for aircraft and related parts. Nowhere is that change seen as revolutionary as the introduction of electronic data and record retention. Just as the arrival of the personal computer placed the possibility of the power and versatility of a computer in the hands of the average person, it made it available to the maintenance technician. Initially some technicians developed their own programs for listing data (TCDS, AD notes, and so forth), but soon commercially available programs were developed. Basically, these were developed by either one of the following two groups:

1. Computer literate persons who felt the aviation industry could benefit from the computer
2. Aviation professionals who felt the aviation industry must benefit from the computer

Some of those initial programs were either not very user friendly (if developed by computer wizards) or not "very sophisticated" (if developed by the maintenance technician). Today, there is a mixture of these various database programs. A review of the advertisement section in any current aviation maintenance magazine offers



UNITED STATES OF AMERICA
DEPARTMENT OF TRANSPORTATION-FEDERAL AVIATION ADMINISTRATION

**STANDARD AIRWORTHINESS CERTIFICATE**

| 1 NATIONALITY AND REGISTRATION MARKS | 2 MANUFACTURER AND MODEL | 3 AIRCRAFT SERIAL NUMBER | 4 CATEGORY |
|---|---|---|---|
| N12345 | Boeing 787 | 43219 | Transport |

5 AUTHORITY AND BASIS FOR ISSUANCE

This airworthiness certificate is issued pursuant to 49 U.S.C. 44704 and certifies that, as of the date of issuance, the aircraft to which issued has been inspected and found to conform to the type certificate therefore, to be in condition for safe operation, and has been shown to meet the requirements of the applicable comprehensive and detailed airworthiness code as provided by Annex 8 to the Convention on International Civil Aviation, except as noted herein.

Exceptions:

None

6 TERMS AND CONDITIONS

Unless sooner surrendered, suspended, revoked, or a termination date is otherwise established by the FAA, this airworthiness certificate is effective as long as the maintenance, preventative maintenance, and alterations are performed in accordance with Parts 21, 43, and 91 of the Federal Aviation Regulations, as appropriate, and the aircraft is registered in the United States.

| DATE OF ISSUANCE | FAA REPRESENTATIVE | DESIGNATION NUMBER |
|---|---|---|
| 9 Jan 2015 | E.R. White *E.R. White* | NE-XX |

Any iteration, reproduction, or misuse of this certificate may be punishable by a fine not exceeding $1,000 or imprisonment not exceeding 3 years or both.
THIS CERTIFICATE MUST BE DISPLAYED IN THE AIRCRAFT IN ACCORDANCE WITH APPLICABLE FEDERAL AVIATION REGULATIONS.

FAA Form 8100-2 (04-11) Supersedes Previous Edition

SAMPLE

**Figure 2-20.** *FAA Form 8100-2, Standard Airworthiness Certificate.*

the reader numerous options for electronic maintenance records. Many of these programs offer a combination of the data research, such as ADs, SBs, STCs, and TCDSs, required to conduct proper maintenance, inspections, and data recording (logbook entries, AD compliance history, length of component time in service, and so forth) desired to improve the efficiency of the technician.

Although some large shops and certified repair stations may have a separate group of people responsible for "records and research," the professional maintenance technician must be aware of the benefits of these systems. Some factors to consider when reviewing a system are:

- What is the typical size of the aircraft that maintenance is being done on? (i.e., less than 12,500 pounds, more than 12,000? Mixed?)
- Does the program have built-in templates for the aircraft being worked on?
- What FAA forms (if any) are available in the program?
- Does it have a user-friendly template to enter the data for the form or must data be directly entered onto the form?
- Can it calculate weight and balance data?
- Does it have adequate word search capabilities?
- Is it networkable?
- Are the updates sent via U.S. mail or downloaded from the Internet?
- What is the maximum number of aircraft that the system can handle?
- Can the system handle both single- and multi-engine aircraft? Fixed and rotary wing? Piston and jet?
- Can an item removed from an aircraft be tracked?
- Is the data from this system exportable to other electronic formats?
- Can it forecast items due for maintenance or inspection?

Since no program can be considered the best, the technician must learn all they can about the numerous systems that exist. Exposure to the pros and cons of these different systems can be one of the benefits of attending various trade shows, maintenance seminars, or IA renewal sessions. Continuous learning and personal improvement is the goal of every professional maintenance technician.

## Light Sport Aircraft (LSA)

### Maintenance

The light sport aircraft (LSA) category includes gliders, airplanes, gyroplanes, powered parachutes, weight-shift and lighter-than-air aircraft. There are two general types of LSAs: Special (SLSA) and Experimental (ELSA). The SLSA are factory built and the ESLA are kit-built. This new category of aircraft was added to the regulations in 2004. (Refer to 14 CFR sections 21.190, 65.107, and 91.327, all dated July 27, 2004.)

## Front

UNITED STATES OF AMERICA
DEPARTMENT OF TRANSPORTATION - FEDERAL AVIATION ADMINISTRATION

**SPECIAL AIRWORTHINESS CERTIFICATE**

| | | | |
|---|---|---|---|
| **A** | CATEGORY/DESIGNATION | | |
| | PURPOSE | | |
| **B** | MANU-FACTURER | NAME | |
| | | ADDRESS | |
| **C** | FLIGHT | FROM | |
| | | TO | |
| **D** | N- | | SERIAL NO. |
| | BUILDER | | MODEL |
| **E** | DATE OF ISSUANCE | | EXPIRY |
| | OPERATING LIMITATIONS DATED | | ARE PART OF THIS CERTIFICATE |
| | SIGNATURE OF FAA REPRESENTATIVE | | DESIGNATION OR OFFICE NO. |

Any alteration, reproduction or misuse of this certificate may be punishable by a fine not exceeding $1,000 or imprisonment not exceeding 3 years, or both. THIS CERTIFICATE MUST BE DISPLAYED IN THE AIRCRAFT IN ACCORDANCE WITH APPLICABLE TITLE 14, CODE OF FEDERAL REGULATIONS (CFR).

FAA Form 8130-7 (04-11) Previous Edition 07/04 May be Used until Depleted *SEE REVERSE SIDE* *NSN: 0052-00-693-4000*

SAMPLE

## Back

| | |
|---|---|
| **A** | This airworthiness certificate is issued under the authority of Public Law 104-6, 49 United States Code (USC) 44704 and Title 14 Code of Federal Regulations (CFR). |
| **B** | The airworthiness certificate authorizes the manufacturer named on the reverse side to conduct production fight tests, and only production flight tests, of aircraft registered in his name. No person may conduct production flight tests under this certificate: (1) Carrying persons or property for compensation or hire: and/or (2) Carrying persons not essential to the purpose of the flight. |
| **C** | This airworthiness certificate authorizes the flight specified on the reverse side for the purpose shown in Block A. |
| **D** | This airworthiness certificate certifies that as of the date of issuance, the aircraft to which issued has been inspected and found to meet the requirements of the applicable CFR. The aircraft does not meet the requirements of the applicable comprehensive and detailed airworthiness code as provided by Annex 8 to the Convention On International Civil Aviation. No person may operate the aircraft described on the reverse side: (1) except in accordance with the applicable CFR and in accordance with conditions and limitations which may be prescribed by the FAA as part of this certificate; (2) over any foreign country without the special permission of that country. |
| **E** | Unless sooner surrendered, suspended, or revoked, this airworthiness certificate is effective for the duration and under the conditions prescribed in 14 CFR, Part 21, Section 21.181 or 21.217. |

**Figure 2-21.** *FAA Form 8130-7, Special Airworthiness Certificate.*

Just as industry standard specifications have replaced many of the military standards to define products that are destined to be part of the Department of Defense (DoD) inventory, so too have industry standards come into the FAA sights for documenting certain information. Quality is one example. The Society of Automotive Engineers (SAE) has developed AS 9100 and AS 9110 as auditing standards for aerospace facilities and specifically repair stations. Likewise, ISO 9001 is being adopted by the FAA as a system of measuring their performance. Therefore, it was logical that when the FAA looked to develop the standards for this newest category of aircraft, they again looked to industry, and this time it was the American Society for Testing and Materials (ASTM).

US Department of Transportation
**Federal Aviation Administration**

# MAJOR REPAIR AND ALTERATION
## (Airframe, Powerplant, Propeller, or Appliance)

OMB No. 2120-0020
Exp: 01/31/2023

Electronic Tracking Number

**For FAA Use Only**

INSTRUCTIONS: Print or type all entries. See Title 14 CFR §43.9, Part 43 Appendix B, and AC 43.9-1 (or subsequent revision thereof) for instructions and disposition of this form. This report is required by law (49 U.S.C. §44701). Failure to report can result in a civil penalty for each such violation. (49 U.S.C. §46301(a))

| | | | |
|---|---|---|---|
| **1. Aircraft** | Nationality and Registration Mark | Serial No. | |
| | Make | Model | Series |
| **2. Owner** | Name *(As shown on registration certificate)* | Address *(As shown on registration certificate)*<br>Address<br>City State<br>Zip Country | |

**3. For FAA Use Only**

SAMPLE

| **4. Type** | | **5. Unit Identification** | | | |
|---|---|---|---|---|---|
| Repair | Alteration | Unit | Make | Model | Serial No. |
| ☐ | ☐ | AIRFRAME | | *(As described in Item 1 above)* | |
| ☐ | ☐ | POWERPLANT | | | |
| ☐ | ☐ | PROPELLER | | | |
| ☐ | ☐ | APPLIANCE | Type<br>Manufacturer | | |

**6. Conformity Statement**

| A. Agency's Name and Address | B. Kind of Agency | | |
|---|---|---|---|
| Name | ☐ U. S. Certificated Mechanic | ☐ Manufacturer | |
| Address | ☐ Foreign Certificated Mechanic | C. Certificate No. | |
| City State | ☐ Certificated Repair Station | | |
| Zip Country | ☐ Certificated Maintenance Organization | | |

D. I certify that the repair and/or alteration made to the unit(s) identified in item 5 above and described on the reverse or attachments hereto have been made in accordance with the requirements of Part 43 of the U.S. Federal Aviation Regulations and that the information furnished herein is true and correct to the best of my knowledge.

| | |
|---|---|
| Extended range fuel per 14 CFR Part 43 App. B ☐ | Signature/Date of Authorized Individual |

**7. Approval for Return to Service**

Pursuant to the authority given persons specified below, the unit identified in item 5 was inspected in the manner prescribed by the Administrator of the Federal Aviation Administration and is ☐ Approved ☐ Rejected

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| BY | ☐ | FAA Flt. Standards Inspector | ☐ Manufacturer | ☐ | Maintenance Organization | ☐ | Persons Approved by Canadian Department of Transport |
| | ☐ | FAA Designee | ☐ Repair Station | ☐ | Inspection Authorization | Other *(Specify)* | |

| | |
|---|---|
| Certificate or Designation No. | Signature/Date of Authorized Individual |

**Figure 2-22.** *FAA Form 337, Major Repair and Alteration.*

NOTICE

*Weight and balance or operating limitation changes shall be entered in the appropriate aircraft record. An alteration must be compatible with all previous alterations to assure continued conformity with the applicable airworthiness requirements.*

**8. Description of Work Accomplished**
*(If more space is required, attach additional sheets. Identify with aircraft nationality and registration mark and date work completed.)*

Nationality and Registration Mark | Date

SAMPLE

Additional Sheets Are Attached

**Figure 2-22.** *FAA Form 337, Major Repair and Alteration (continued).*

The ASTM developed a comprehensive list of consensus standards for use by manufacturers, regulators, maintenance facilities, LSA owners, and service providers. It is unique that these standards are the first ones in over 100 years to solely address the issue of recreational aircraft use. It is also the first complete set of industry consensus standards covering the design, manufacture, and use of recreational aircraft that was developed by a non-government agency. The ASTM committee that developed these LSA standards did so to ensure the quality of products and services to support both the national and the international regulatory structures for LSAs. Over 20 standards have been generated, and more are being developed to cover this diversity of aircraft. This handbook only incorporates a review of F2483-05, "Standard Practice for Maintenance and the Development of Maintenance Manuals for Light Sport Aircraft (LSA)" a six-page document comprised of the following 12 sections:

1. Scope
2. Referenced Documents
3. Terminology
4. Significance and Use
5. Aircraft Maintenance Manual
6. Line Maintenance, Repairs, and Alterations
7. Heavy Maintenance, Repairs, and Alterations
8. Overhaul
9. Major Repairs and Alterations
10. Task-Specific Training
11. Safety Directives
12. Keywords

The scope of that document is basically twofold:

- To provide guidelines for the qualification necessary to accomplish various levels of maintenance on LSA.
- To provide the content and structure of maintenance manuals for aircraft and their components that are operated as LSAs.

Some additional definitions from section 3, Terminology, that help to better explain the LSA concepts are:

- Annual condition inspection—defined as a detailed inspection accomplished once a year in accordance with instructions provided in the maintenance manual supplied with the LSA. The purpose of this inspection is to look for any wear, corrosion, or damage that would cause the LSA not to be in condition for safe operation.
- Heavy maintenance—any maintenance, inspection, repair, or alteration a manufacturer has designated that requires specialized training, equipment, or facilities.
- Line maintenance—any repair, maintenance, scheduled checks, servicing, inspections, or alterations not considered heavy maintenance that are approved by the manufacturer and is specified in the manufacturer's maintenance manual.
- LSA repairman–inspection—a U.S. FAA-certified LSA repairman with an inspection rating per 14 CFR part 65. This person is authorized to perform the 100-hour/annual inspection of the aircraft that they own.
- LSA repairman–maintenance—a U.S. FAA-certified LSA repairman with a maintenance rating per 14 CFR part 65. This person is allowed to perform the required maintenance and can also accomplish the 100-hour/annual inspection.
- Major repair, alteration, or maintenance—any repair, alteration, or maintenance where instructions to complete the task are excluded from the maintenance manual.
- Minor repair, alteration, or maintenance—any repair, alteration, or maintenance where instructions to complete the task are included in the maintenance manual.

The 100-hour inspection is the same as the annual inspection, except for the interval of time. The requirements for whether or not the 100-hour inspection is applicable are exactly the same as the criteria for the standard 100-hour/annual required of non-LSA aircraft.

### Aircraft Maintenance Manual (AMM)

Although these manuals do not require any FAA approval, the regulations do require that the manual be developed in accordance with industry standards. This ASTM sets that standard by requiring:

- General specifications to be listed, include capacities, servicing, lubrication, and ground handling
- An inspection checklist for the annual condition or 100-hour inspection
- A description of and the instructions for the maintenance, repair, and overhaul of the LSA engine
- A description of and the instructions for the maintenance, repair, and alteration of the aircraft's primary structure

Other items that maintenance procedures must be provided for are:

- Fuel systems

- Propeller
- Utility system
- Instruments and avionics
- Electrical system
- Structural repair
- Painting and coatings

The Inspection, Repair, and Alterations section must specifically list any special tools and parts needed to complete the task, as well as the type of maintenance action (line, heavy, or overhaul) necessary to accomplish the activity. Directly associated with that information is the requirement to specify the level of certification needed to do the job (i.e., LSA repairman, A&P, or repair station). The manual may refer to existing FAA ACs.

### Line Maintenance, Repairs, & Alterations

The minimum level of certification necessary to accomplish line maintenance is LSA inspection. Some typical tasks considered to be line maintenance are:

- 100-hour/annual condition inspection
- Servicing of fluids
- Removing and replacing components when instructions to do so are provided in the maintenance manual
  - Batteries
  - Fuel pump
  - Exhaust
  - Spark plugs and wires
  - Floats and skis
- Repair or alteration of components when specific instructions are provided in the maintenance manual
  - Patching a hole in the fabric
  - Installation of a strobe light kit

Heavy maintenance, repairs, and alterations must be accomplished by either a certified mechanic (A or P or A&P) or an LSA repairman—maintenance who has received additional "task specific" training. Some examples of this would be the removal and replacement of complete engine, cylinder, piston and valve assemblies; primary flight controls; and landing gear.

Heavy repair of components or structure can be accomplished when instructions are provided in the maintenance manual or other service directed instructions. A few examples of this activity are:

- Repainting of control surfaces
- Structural repairs
- Recovering of a dope and fabric

Heavy alterations of components can be accomplished when instructions are provided in the maintenance manual or other service directed instructions. Examples of this activity are initial installation of skis and installation of new additional pitot static instruments.

Overhaul of components can be performed only by the manufacturer (or someone authorized to perform) of the LSA or the component to be overhauled. An overhaul manual is required and must be a separate manual from the manufacturer's maintenance manual. Items typically considered for overhaul are engines, carburetors, starters, generators, alternators, and instruments.

### Major Repairs & Alterations

Another major difference between LSA maintenance and traditional aircraft maintenance is that FAA Form 337, Major Repair and Alteration, is not required to document major repairs and alterations. Instead, any major repair or alteration that is accomplished after the LSA has gone through production acceptance testing must be evaluated relative to the applicable ASTM requirements. After this evaluation has been accomplished (either by the manufacturer or an entity approved by them), a written affidavit must be provided attesting that the LSA still meets the requirements of the applicable ASTMs.

The manufacturer (or other approved entity) must provide written instructions defining the level of certification necessary to perform the maintenance and also include any ground test or flight testing necessary to verify that the LSA complies with the original LSA acceptance test standards, and is in condition for safe operation. Proper documentation of this maintenance activity is required to be entered in the LSA records and is also defined by the manufacturer.

Task specific training is not required to be FAA approved. This is solely the responsibility of the manufacturer. Some examples of this are an engine manufacturer's overhaul school or the EAA Sport Air fabric covering school.

Safety directives are issued against an LSA or component and are not issued by the FAA, but rather by the original aircraft manufacturer. ***Note:*** If the LSA includes a product that is TC'd by the FAA, the manufacturer is required to issue a safety directive. Typical instructions within a safety directive include:

- List of tools required for the task
- List of parts needed

- Type of maintenance (line, heavy, overhaul)
- Level of certification needed
- Detailed instructions and diagrams
- Inspection and test methods

Safety directives are mandatory, except for experimental use LSAs.